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A Neural Network Framework for Estimation of Three Dimensional Object Location in Small Rooms Using Acoustic Echoes

A thesis submitted in partial fulfilment to opt for the degree of

Magister Scientiæ in Computer Science

Author
Esteban Roberto Padilla Padilla

Supervisor
Ing. Kervin Sánchez Herrera, M.Sc.

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Thesis approval certificate goes in here.

To my wife Rebeca and my family, for the time they granted me and for their unwavering support throughout the writing of this thesis.

A mi esposa Rebeca y a mi familia, por el tiempo cedido y por su apoyo durante la realización de esta tesis.

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Abstract

This document presents the thesis for obtaining the Master's degree in Computer Science: Emphasis in Computer Science at the Tecnológico de Costa Rica. The topic is centered in the idea of the use of machine-learning techniques to learn patterns from acoustic echoes produced inside small rooms and estimate the location of objects in three-dimensional space. Two neural-network architectures are employed for this task: a Convolutional Neural Network (CNN), which, although primarily designed for image processing, is commonly used in the literature for acoustic inference; and an Echo State Network (ESN), a memory- and computationally efficient architecture with the potential to process acoustic signals more directly. The effect of beamforming is also investigated as a possible solution to multipath acoustic interference. The network designs, experimental setup, and methodology for measuring, processing, and evaluating the results are presented to assess the potential of audible-frequency sound echoes for locating objects in an enclosed room.

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Chapter 1

Introduction

Robots are expected to navigate multiple and diverse environments. To achieve this, they need situational awareness and the autonomy to make proper decisions. Computer vision aims to enable machines to interpret the world that surrounds them. By processing images and videos, robots can extract information to recognize object geometries, textures, materials, etc. Although visual analysis allows a machine to build internal models containing vast knowledge of its environment, this does not mean that its perception is complete. Robotic perception gives a machine the ability to understand its surroundings to the degree that it can safely interact with the world [17]. Such understanding requires other techniques in addition to computer vision to obtain a complete sensory experience of the environment. Acoustic sensing and analysis can enhance machine perception by following an approach similar to that seen in nature with animals that use binaural acoustic perception: the environment can be analyzed to detect the source of a particular sound and focus visual attention on that region. However, this capability is not only an extension of vision. In some conditions, animals rely entirely on acoustic information to operate, as in the

echolocation employed by bats, birds, dolphins, and whales, which hear the echoes of acoustic chirps to gain spatial perception of their surroundings and navigate complex layouts [49].

Traditional computer vision techniques that employ sensing devices such as cameras and lidars provide high-fidelity information to navigation systems, but they become unreliable in scenarios with poor lighting or the presence of dust, smoke, and fog [42]. Another aspect to consider is the cost of the sensing device. Acoustic perception offers an advantage in this regard due to the low cost of commercial electret or MEMS microphones compared to optical cameras and lidars, which makes it desirable as an alternative to traditional sensors. Recent developments in machine learning, such as those shown in [7; 35; 42], have demonstrated the great potential of neural networks to infer spatial and geometrical information about surrounding objects by analyzing acoustic echoes in the audible spectrum. The manufacturing sector, under the Industry 4.0 transformation, is increasingly adopting automation and smart monitoring technologies, which makes it an important field of application for these new methods. This is especially true for the supervision of machinery during operation, particularly machines on production floors that use robotic arms and computer numerical control actuators to manipulate objects in confined spaces [31]. In such conditions, acoustic sensing is a good approach to gain insight into the overall status of the system, because the Room Impulse Response acts as a transfer function that encodes information about the environment setup. The acoustic response of a room can be used together with artificial neural networks, such as convolutional neural networks or recurrent neural networks, to learn from this response and recognize characteristics and properties of the room echoes, and thereby recognize spaces and objects within it. Thus, acoustic perception together with neural networks can improve the perception of machinery operating in complex spaces.

1.1. Problem

The use of acoustic signals to approximate the size and shape of objects has been considerably improved by the adoption of machine learning techniques. An important part of the work in this field was developed for indoor navigation systems, where the goal is to enhance the perception of robots by using acoustic analysis to build low-resolution depth images of an unknown environment. This work presents a different point of view. First, the environment is completely known a priori, but an object changes its position inside it, and its location must be estimated with a certain accuracy each time. Second, in contrast to the large rooms in which those systems are intended to navigate, the object location must be estimated at short ranges, where multi-path echoes can be received at the same time. This creates a superposition that is difficult to resolve and can cause those systems to malfunction [7]. To the best of our knowledge, no previous work has estimated an object's position using acoustic analysis under the conditions described. To achieve acoustic spatial estimation in this scenario, we propose two changes to the processing pipeline. The first is in the beamforming stage, to increase the spatial discrimination of the origin of the echoes. The second is a change in the type of neural network used to infer the location, employing a network suited to directly process sequences of acoustic features instead of the images required by Convolutional Neural Networks. Our purpose is to demonstrate that it is possible to locate an object in a small room (e.g., 1 m side length) using acoustic signal processing with neural networks.

1.2. Document Structure

This document present in **Chapter 1** a brief introduction to the topic and is stated the problem we intent to address with the development of this thesis. **Chapter 2** contains the theoretical background and state of the art in the field. **Chapter 3**

details the hypothesis, objectives and scope of the work. **Chapter 4** shows the research proposal and methodology. **Chapter 5** present the results obtained from the experiments conducted. **Chapter 6** discusses the findings and their implications. **Chapter 7** concludes the thesis and outlines future work.

Chapter 2

Theoretical Framework

In this chapter is described the theoretical concepts needed to understand the work developed during the research's execution.

2.1. Room Impulse Response

An acoustic impulse is a sharp, narrow pulse whose spectrum is composed of a uniform distribution of all frequencies. The mathematical Dirac delta function $\delta(t)$ is used to represent this kind of impulse. In enclosed spaces, the sound waves generated by an impulse travel, reflect, and refract on the walls delimiting the space. A microphone measuring the environment first receives the direct-path sound, which travels from the source directly to the microphone, followed by the early reflections and finally by a series of pulses called late-field reverberation, as shown in Figure 2.1. The direct path, early reflections, and reverberation correspond to the response of the enclosed space to an acoustic impulse and are physically defined by the geometry and materials of the environment [39].

The Room Impulse Response (RIR) is, by definition, the transfer function of a sound wave that travels through the space from its source to the microphone. A microphone records the convolution of the emitted sound with the corresponding RIR [15], as shown mathematically in Equation 1 with $g(t)$ representing the recorded signal, $h(t)$ the RIR, and $x(t)$ the emitted sound. Therefore, any sound generated at one point and received at another inside a room contains the effects of the RIR of the environment.

$$g(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(t)x(t-s)ds \tag{1}$$

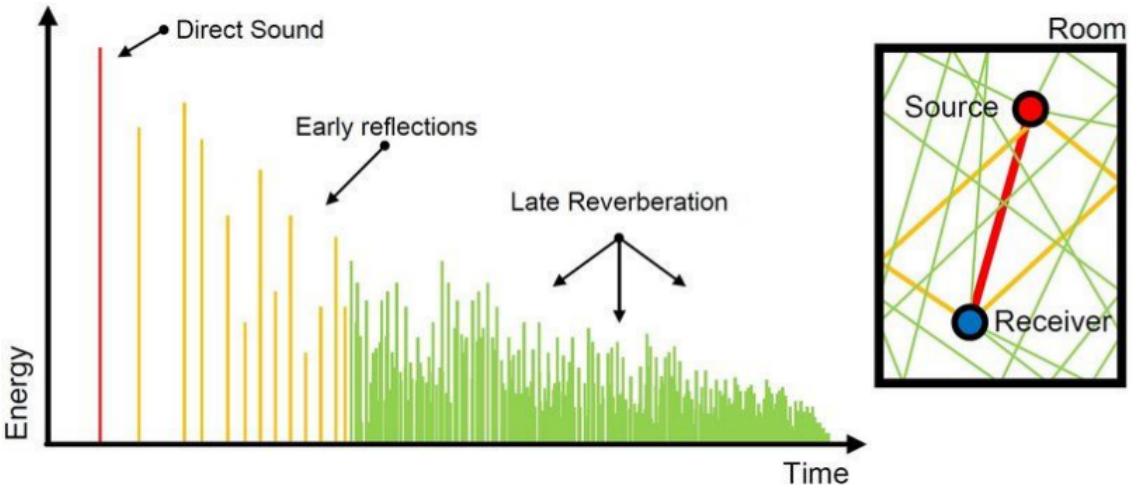


Figure 2.1: Room Impulse Response and the geometrical room representation [36].

2.2. Acoustic Beamforming

In many spatial omnidirectional data transfer contexts, not only the information carried in the medium is important: the source from which the signal is generated also provides valuable information for properly analyzing the incoming data. Beamforming is a spatial filtering technique that aims to estimate the direction of arrival of a signal,

even in the presence of noise and other interference in the propagation medium [11]. In acoustics, an array of microphones is used to capture spatially propagated signals arriving from a certain direction and to process them in order to investigate the acoustic sources [29].

2.2.1. Microphone Array

A microphone array combines the signals of multiple omnidirectional microphones in such a way that it captures sound from a specific direction while rejecting sound from other directions [47]. The shape and dimensionality of the arrangement depend on the application. The simplest form consists of two microphones separated by a distance D, as shown in Figure 2.2. The value of D is the minimum distance for a given frequency at the sound speed c, and its calculation is shown in Equation 2. When the input signals are simply added together for a given frequency f, this arrangement captures sound coming from ± 90 degrees and rejects sound arriving from 0 and 180 degrees. In a microphone array, a higher number of microphones results in greater gain in the desired direction as well as greater attenuation of sound coming from the sides of the array [38].

$$D = \frac{1}{4}\frac{c}{f} \tag{2}$$

2.2.2. Beamforming Pipe

2.2.2.1. Delay-and-Sum Beamforming Technique

Beamforming is based on the physical principle that an acoustic wave originating at a particular location in space travels and arrives at each sensor at a different time, defined by the distance between the single source and each sensor. The result of adding all these signals depends on the phase relationship between them, so it is possible to

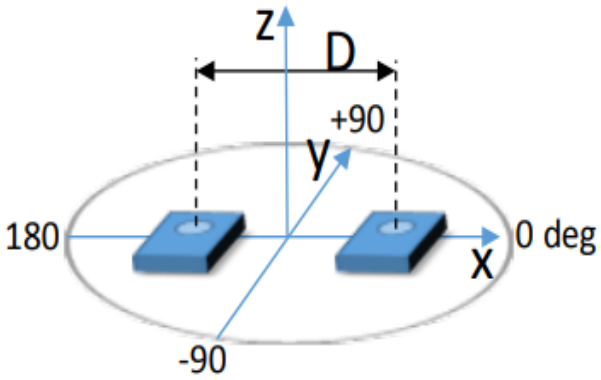


Figure 2.2: Microphone array in binaural arrangement [47].

control the delay of each signal to constructively add those from a particular location while, at the same time, destructively canceling those originating in other regions [11]. Mathematically, the beamforming formula is shown in Equation 3, where $L(t, S_0)$ is the beamforming output, $p_m(t + \tau_m(\mathbf{s}_0))$ is the acoustic pressure signal recorded by microphone m, advanced by the specific time-of-flight delay (τ_m or t_0) it takes for sound to travel from the coordinate $\mathbf{s}_0$ to that specific microphone, and $\|\mathbf{s}_m - \mathbf{s}_0\|$ is the spherical spreading compensation factor. Because acoustic energy decays proportionally to the distance traveled ($1/r$). All the signals are summed together and divided by the total number of microphones M in the array.

$$L(t, \mathbf{s}_0) = \frac{4\pi}{M} \sum_{m=1}^{M} p_m(t + \tau_m(\mathbf{s}_0)) \|\mathbf{s}_m - \mathbf{s}_0\| \tag{3}$$

2.3. Time Series Transformation into Images

Given a time series, it is sometimes desirable to apply certain transformations that provide a more compact representation of the data or highlight specific properties and attributes of it. Transforming a time series into an image involves converting vectors

into matrices using a transformation criterion that maps values from one domain to another. One example is the conversion of audio signals into spectrograms or Log-Mel spectrograms, which provide a time-frequency representation of the signal. In machine learning applications, these image-like representations can be used as input to convolutional neural networks and other image-based models, enabling the extraction of meaningful features from time series data.

2.3.1. Log-Mel Spectrogram

Spectrograms are widely used in audio and speech processing, as well as in other fields where time-varying signals are analyzed [3]. A spectrogram is a visual representation of the spectrum of frequencies of a signal over time. It is obtained by dividing the signal into overlapping segments, applying a window function to each segment, and computing the Fourier transform to obtain the frequency content. The resulting magnitude is then plotted as a function of time and frequency, providing a time-frequency representation of the signal.

2.3.2. Log-Mel-Scale

Pitch, in acoustics, is the perceptual quality that allows sounds to be ordered from low to high by frequency, which is what the human mind does when processing tones [18]. The Mel scale is a representation of how humans perceive audio pitches. It is based on the ability of humans to judge equal distances between pitches.

The conversion from frequency to the mel scale is given by Equation 2.1.

$$m(f) = 2595 \log_{10}\left(1 + \frac{f}{700}\right) = 1127 \ln\left(1 + \frac{f}{700}\right) \tag{2.1}$$

For the discretization of the mel frequencies, the continuous mel scale is filtered using a bank of triangular bandpass filters, each representing a specific range of mel

values. The canonical number of filters is 128 in Machine Learning applications with complex environmental audio tasks, in order to preserve fine spectral cues [18].

2.4. Convolutional Neural Network

A Convolutional Neural Network (CNN) is an artificial neural network architecture used to analyze and learn from visual data, commonly presented in the form of images. In its simplest form, the structure is composed of three building blocks: the convolution layer, the pooling layer, and the fully connected layer. Figure 2.5 shows an example of how these blocks are connected to analyze an image stage by stage in order to determine the structural condition of an aircraft fuselage.

2.4.1. Convolution operation

Convolution is a common operation used in image filtering to denoise or detect features. A filter, also called a convolution kernel, is applied to all locations of the image to produce a new filtered image [14]. A filter W of dimension $K \times L$ with weights $w_{k,l}$ can be convolved with a given 2D image X of dimension $I \times J$ with pixels $x_{i,j}$ to obtain a filtered image $X * W$. The convolution process over the image can be mathematically written as:

$$(w * x)_{i,j} = \sum_{k=-K/2}^{K/2} \sum_{l=-L/2}^{L/2} w_{k,l} x_{i+k,j+l} \tag{5}$$

The convolution operation in Equation 5 does not allow the center of the kernel to overlap the edges of the image, resulting in a reduction in the size of the output feature map. Padding increases the input size by adding extra rows and columns around the outermost elements on each side of the image in order to compensate for

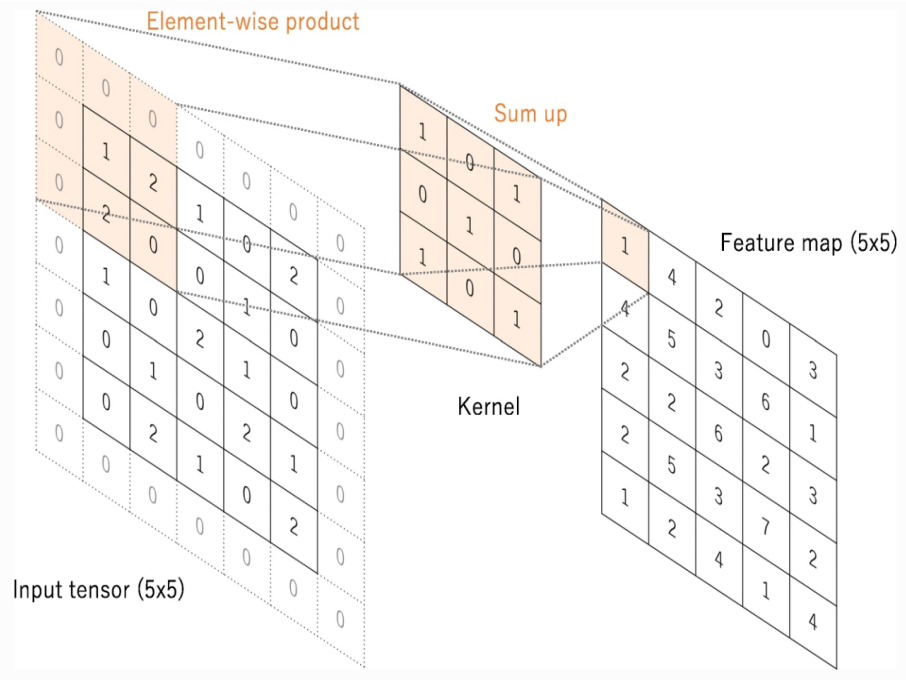


Figure 2.3: Convolution operation over a tensor of 5x5 with zero padding and a stride of 1, resulting in a feature map of 5x5 [51].

this dimensional reduction [51]. In practice, as shown in Figure 2.3, the kernel is applied across the input one step at a time. An element-wise operation is performed at each location and the results are summed to obtain the output. The kernel is then moved to the next location to filter another segment of the input. The size of the kernel displacement is given by the stride, which defines the distance between two successive kernel positions.

2.4.2. Pooling

Pooling is a dimension reduction technique in which an input map is spatially reduced while keeping the most important features that were filtered or learned. The pooling operation reduces or collapses a subregion to a smaller dimension, helping the learned representation become more invariant to small length-scale perturbations [22].

Two common pooling techniques are max-pooling and average-pooling. For a defined pooling window, max-pooling considers only the maximum value, whereas average-pooling calculates the average of the values inside the window, as shown in the example in Figure 2.4.

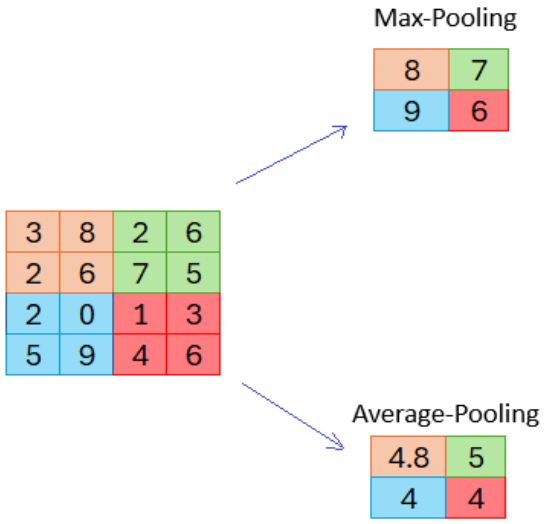


Figure 2.4: Pooling escalation with average pooling and max-pooling.

2.4.3. Fully connected layer

The final stage in a CNN is a fully connected layer. This layer is responsible for classifying the features extracted by the previous layers. Figure 2.5 shows an example of a CNN for aircraft structural health classification, with its fully connected layer located at the output of the network.

2.5. Echo State Network

An Echo State Network (ESN) is a recent paradigm in recurrent neural networks known as Reservoir Computing (RC). In RC, the reservoir is the recurrent part of the

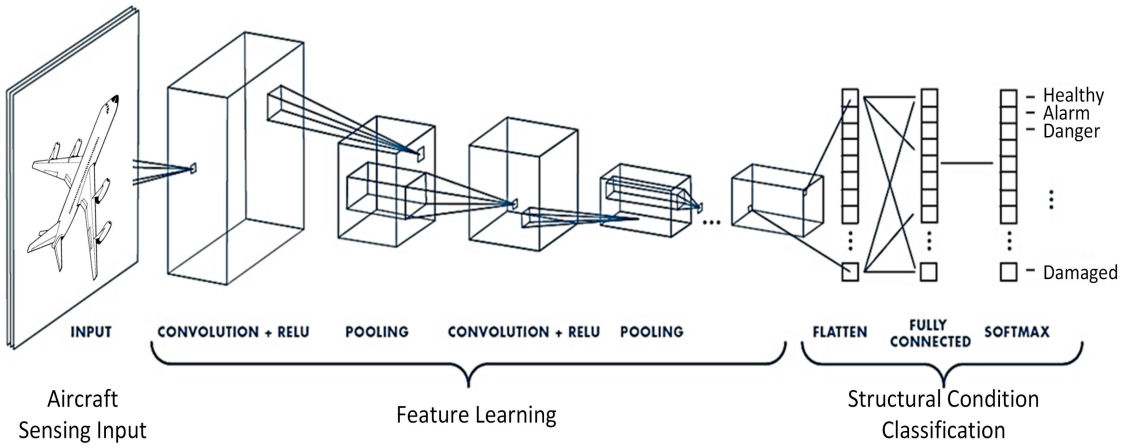


Figure 2.5: Architecture of Convolutional Neural Network for aircraft structural health monitoring [41].

network and has the particularity of not being trained. Because the reservoir skips training, the network avoids the computationally expensive gradient-descent training of recurrent networks. Instead, the recurrent network is defined in such a way that it can transform input signals into values in a high-dimensional state space. The state of the reservoir can be processed by a simple fully connected readout network trained to map the reservoir state to a desired output [26].

In the ESN, the reservoir is a randomly created RNN whose weights remain unchanged. This RNN is excited by an input signal, and its future state is a function of the current input and the history of the reservoir. The result is generated by training a simple feed-forward network to output the target value. Figure 2.6 [26] shows a comparison of a traditional RNN and RC. In RC, the bold arrows exemplify the weight changes resulting from the training process, which is performed only at the output layer of the network.

2.5.1. Formalism of ESN

A formal description of an ESN, given in [40], defines an ESN that receives a time series input $U(t) \in R^{D\times 1}$, where the state of the reservoir is denoted by $X(t) \in R^{N\times 1}$

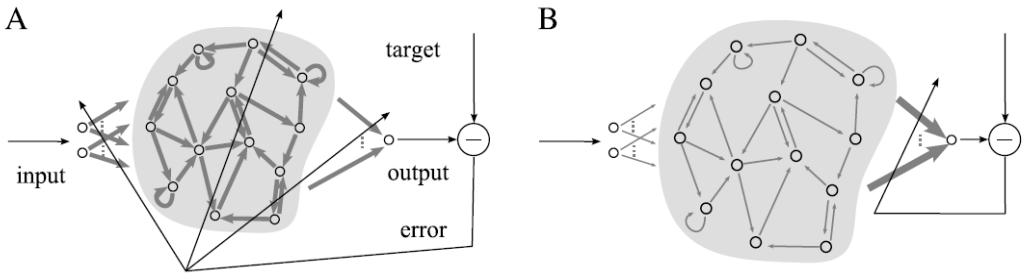


Figure 2.6: Comparison of B) RC vrs A) traditional RNN, the bold arrows are weights that change during training [26].

and the output by $y(t) \in R^{M\times 1}$. The weights for the stages are $W_{in} \in R^{N\times D}$ for the input, $W_{res} \in R^{N\times N}$ for the reservoir, and $W_{out} \in R^{N\times M}$ for the output. The equations for the internal state of the reservoir and the output are shown in Equation 6. The reservoir state depends on the input value at time t and the previous state of the system. The output can be obtained by training to solve a linear regression problem.

$$\begin{aligned} x(t) &= f(W_{in}U(t) + W_{res}x(t-1)) \\ y(t) &= W_{out}x(t) \end{aligned} \tag{6}$$

There is an important property that the reservoir should meet to ensure that its dynamics are capable of forgetting the initial transient state [50]. The Echo State Property (ESP) states that a network with state transition function F has the echo state property if, for each input sequence $U = [u(1), \ldots, u(N)]$ and all pairs of initial reservoir states x, x', Equation 7 holds.

$$||F(U, x) - F(U, x')|| \to 0 \; as \; N \to \infty \tag{7}$$

To ensure the existence of the ESP, W_{res} must meet a necessary condition and a

sufficient condition, defined as follows:

$$\begin{aligned} ESP &\Rightarrow \rho(W_{res}) < 1 \\ \bar{\sigma}(W_{res}) < 1 &\Rightarrow ESP \end{aligned} \tag{8}$$

where $\rho(W_{res})$ is the spectral radius of W_{res} and $\bar{\sigma}(W_{res})$ is its largest singular value. Thus, from Equation 8, if $\rho(W_{res})$ and $\bar{\sigma}(W_{res})$ are less than one, the ESP is satisfied and the reservoir dynamics will behave correctly.

2.6. Related Work

The related work in acoustic processing for object recognition and classification can be divided into two categories based on where the sound originates. An approach is considered passive sensing if the sound stimulus originates directly in the system. Conversely, if the sound is an external stimulus that interacts with the environment before being sensed, the approach is active sensing.

2.6.1. Passive Acoustic Sensing and Analysis

2.6.1.1. Acoustic Imaging

An acoustic image is a sound intensity map that highlights the location of different sound sources in a two-dimensional plane. An acoustic camera takes this intensity map and superimposes it on an optical image to show the physical source of the sound. As shown in Figure 2.7, a microphone array is used to capture the sound, and a beamforming algorithm processes the signals to locate the direction from which the sound originates.

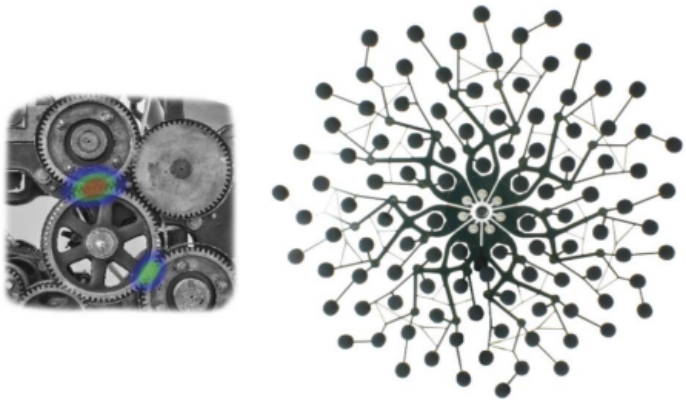


Figure 2.7: Image on the left is an acoustic image, on the right and microphone camera used to capture the image [37].

Acoustic beamforming is an extensively studied field. Different shapes and sizes of microphone arrays have been analyzed [38], and many techniques exist for beamforming the signals from the array [29]. Recent studies have focused on spherical arrays to identify multiple sound sources in three-dimensional space [12; 20; 24].

2.6.1.2. Acoustic Scene Classification

Artificial Intelligence, and especially its deep learning subfield, has powered the development of new techniques to segment, analyze, and classify acoustically originated signals. Acoustic Scene Classification (ASC) uses deep learning to learn and distinguish, from audio, the context in which a recording was made. CNNs are the first choice for ASC. The images required as input to the network are obtained by processing an audio dataset to convert it into spectrograms. Using the DCASE 2016 dataset, the authors of [10] used static log-mel spectrograms and dynamic first-derivative spectrograms to build the two channels of the images used as input for training the CNN. Han et al. in [19] used the DCASE 2017 dataset and improved the training input data through multiple versions of mel spectrograms based on binaural representation, source separation, and background subtraction. In the binaural

representation, left-right and mid-side spectrograms were used instead of a mono audio representation. Source separation divides the dataset in two using harmonic-percussive sound separation algorithms, which increases the dataset by providing harmonic and percussive representations of each sound. Finally, median filtering is applied to the spectrograms to suppress steady environmental noise. The resulting multiple spectrogram versions were used to train a ConvNet architecture of eight convolution layers inspired by a VGG network. Compared with the baseline used in that work, the authors obtained an improvement in mean accuracy, from 74.8% to 80.5% with the various preprocessing methods described above. The work presented in [44] used the DCASE 2019 dataset and log-mel spectrograms to train two deep learning algorithms to perform the classification: a CNN and a Convolutional Recurrent Neural Network (CRNN). The highest recognition accuracy was obtained by the CRNN, with 90.96%, a value superior to the 73.61% of the CNN used as the baseline. A different objective is presented by Byttebier in [8], where small-footprint neural network models are investigated. The authors compared several small neural networks using the DCASE 2020 dataset and found that a 16-bit CNN with squeeze-excitation blocks, regularized with mixup and weight decay, shows the best accuracy, 66.84%, with only 90 kilobytes of parameters.

2.6.1.3. Acoustic Object Recognition

Acoustic object recognition aims to recognize an object based on the sound it emits, whether the sound is self-generated, such as the sound produced by an air conditioner during normal operation, or external, produced in response to a stimulus, such as the characteristic sound of wood when someone knocks on a door. In the work presented by Luo et al. in [27], a dataset was built using 30 different objects, each of which was hit with a marker pen to generate a characteristic sound that was recorded. A training set of 100 hits per object (at different locations on the object) was created, and an extra

20 hits per object were obtained for the test phase. With this dataset, a denoising autoencoder was trained to recognize the different sounds of the objects from which they originate. The recognition rate for the autoencoders was 91.50%, a value superior to the baseline provided by a Support Vector Machine with a recognition rate of 82%. Other dataset was created by Tran et al. in [43], in this case the input were LEGO objects that fall from 30cm high. The sound of the LEGO pieces hitting the ground was used to create spectrograms to training a CNN called LegoNet in order to recognize the LEGO pieces by the sound of its free-fall impact, LegoNet achieved an accuracy of 99.21%, a value superior to the other networks used for comparison. For robotic perception in [23] was develop a framework in which a robot can recognize the object inside a container by shaking the container and analyzing the sound. An kernel K-nearest neighbor algorithm in a open environment is used to shows the feasibility of the method. A generative physical, audio-visual synthesis pipeline is presented in [52], the model takes as input the shape of an object, its material and one scene configuration, a physical engine simulate motion and collision for that particular object. An audio engine takes the shape, material and the collision information to synthesize the sound. The database generated is used in the training a two-stream CNN, one ResNet-18 for the image data, together with a SoundNet-8 for the audio data. The net demonstrated that the knowledge learned in the synthetic audio-visual dataset was transferred to real world data once it was used to infer object materials from auditory data, and estimate shape attributes from audio-visual data.

2.6.2. Active Acoustic Sensing and Analysis

2.6.2.1. RIR for Room Inference

For an enclosed space, the RIR provides a unique representation of the geometry and materials inside it. Sounds such as music and speech recorded in rooms carry this echo information and can be used to recognize the room in which the recording

was made. This idea is presented in [33], where the authors developed a machine learning system to identify rooms from reverberant audio samples recorded in them and achieved 85% accuracy in the task. The work in [6] uses the RIR and speech data to train a model to infer the room volume from sound recorded with a single microphone. With this information, the model classified rooms into three categories (small, large, or hall) with an accuracy of 85.8%. Using microphone arrays and analyzing the RIR obtained from them, the authors of [16] were able to estimate the room geometry by exploring the properties of Euclidean distance matrices, and showed that under certain conditions the first-order echoes provide a unique description of the geometry of convex polyhedral rooms. In [25], an indoor positioning device is presented that uses the RIR as a "fingerprint" to identify the room in which the system is located. The accuracy varies depending on the type of room, ranging from 65% for living rooms to 95% for bathrooms. The work in [48] presents a dataset of RIRs in which people stood in rooms and were relocated to different positions inside them to generate samples. The authors show the potential of using machine learning approaches on the dataset for tasks related to localizing, identifying, and detecting humans in those rooms.

2.6.2.2. Acoustic Object Shape Estimation

Acoustic perception of shapes is explored in [28]. This work proposes the use of acoustic standing waves to recognize the size and shape of a target object by analyzing the distance spectrum profile of the standing waves created by the interference between transmitted and reflected sounds. In this work, Manabe et al. were able to estimate the dimensions of a rectangular board of 1.2 m $\times$ 0.9 m, with a mean absolute error of 0.0194 m. With this result, they demonstrated that the spectrum profile encodes information about the target and can be used to estimate the shape of an object. The approach presented by Brunetto et al. in [7] resulted in the development of the Batvision system, which is capable of seeing depth and spatial

layouts by listening to echoes recorded by two microphones acting as artificial ears. For Batvision, a dataset was created using mobile robots that generate acoustic chirp signals at the same time a depth image is captured. With this dataset of audio-image pairs, an adversarial discriminator was trained to generate depth images based on sound chirps. The system was capable of predicting detailed indoor scenes and obstacles with an L1 loss as low as 0.072, sometimes even outperforming a depth map created by a stereo vision algorithm. Following the findings in Batvision, the work presented in the CatChatter paper [42] shows two separate results. First, an improved version of Batvision was developed in which an input encoder was removed from the model. This encoder processed the acoustic spectrogram before it reached the adversarial discriminator and caused spatial information from the spectrogram to be lost by the time it reached the U-Net, decreasing the inference capacity of the model. The second result presented in [42] was a new system that uses lidar to create 3D, 360-degree depth images, together with an increase in acoustic perception from two microphones to four in order to capture echoes from all directions. The results obtained by this new method are similar to those reported for Batvision [7], but thanks to the 360-degree depth information and the two extra microphones, the field of prediction was four times larger than that obtained with the depth camera and two-microphone system. To detect basic shapes and deformations, acoustic sensing was used in [35] by Rothstein et al. to identify and monitor small hollow objects from the inside. The deformations of the shapes were created by generating similar shapes with altered dimensions or by blocking a certain part of the object's interior. A square wave sound signal is emitted inside the objects, and a four-microphone array records the echoes produced in the interior. Acoustic features are created from these signals, and a classifier is trained to distinguish the shape and deformation of the object under study. Their work showed that, for the objects tested, the shape can be identified with 80% accuracy, the existence of a deformation with 100% accuracy, and

the location of the deformation with 90% accuracy.

Chapter 3

Hypothesis and Objectives

3.1. Hypothesis

The studies reviewed in Chapter 2 demonstrate that sound sources can be located in three-dimensional space with some precision, also that rooms can be recognized by their acoustic signatures, depth images of parts of a room can be created using only audible sound echoes, and object shapes can be inferred from acoustic features extracted from echoes originating inside the objects. Collectively, these findings suggest that a function can be learned to map the RIR encoded in acoustic echoes to the location of an object in an enclosed space. Based on these findings, our hypothesis is:

"An algorithm pipeline with beamforming and ESN that utilize the rich RIR spatial-geometrical information comprehended in a set of acoustic signals can infer the location and orientation of a cube of 20cm side allocated inside an small enclosed space of near of $1m^3$ [1], this with a movement distance resolution of at least 5cm and achieve less

[1] Region with acoustic superposition found by [7]

than 2cm of Mean Absolute Error in position and 10 degrees in orientation[2] with an error less than 6 degrees of Mean Absolute Error in the estimation from two consecutive positions."

3.2. Objectives

The current research established the following objectives:

3.2.1. Main Objective

To model and evaluate a processing pipeline that combines beamforming with CNN and ESN to map multichannel acoustic signals to an object's spatial coordinates and orientation within an enclosed space.

3.2.2. Specific Objectives

1. Collect a dataset of acoustic responses from an object at different locations in a room, together with the corresponding spatial coordinates and orientations.

2. Develop a CNN-based reference method that learns to estimate an object's coordinates and orientation from the dataset.

3. Develop a resource efficient ESN based method that learns to estimate an object's coordinates and orientation from the dataset.

4. Build a beamforming stage that divides the measurement space into four regions based on acoustic source locations.

5. Experimentally evaluate the accuracy of the estimators.

[2]Based on the size of the objects and maximum deformation presented in [35]

3.2.3. Deliverables

The following deliverables are associated with each specific objective defined for the thesis:

Specific Objective 1: A dataset containing acoustic recordings of echoes in the form of time series and its corresponding object location-orientation.

Specific Objective 2: Thesis document with detailed description of the concepts, observation and results analysis that justify the system design with CNN.

Specific Objective 3: Thesis document with detailed description of the concepts, observation and results analysis that justify the system design with ESN.

Specific Objective 4: Thesis document with detailed description of the concepts, observation and results analysis that justify the beamforming.

Specific Objective 5: Thesis document with analysis and evaluation of the proposed solutions considering performance metrics under experimental operation.

3.3. Scope and Limitations

Scope and limitation 1: The development of the measure box and the collection of the dataset is part of the work.

Scope and limitation 2: Only two ML approaches are considered: CNN as the ground truth and ESN as an efficient solution.

Scope and limitation 3: The proposed pipelines will not work on real time, even for the experimental evaluation the data will be first collected and later presented to the model in order to obtain the evaluation metrics.

Scope and limitation 4: The programming language is python, the hardware to process the data will be a desktop personal computer.

Chapter 4

Methodology

The following chapter details the methodology used in this research, including the experimental setup, data collection process, the processing pipeline for acoustic measurements, the models metrics used for location-orientation estimation, and the evaluation criteria for the proposed solutions.

4.1. Experimental measurement configuration

The experimental container is a cubic box with a side length of 1 m and a volume close to $1\,\mathrm{m}^3$. The concept is shown in Figure 4.1(a). Three of its six walls are made of porcelain tiles with a thickness of 1 cm, one wall and the floor are made of solid concrete, and the remaining wall used for observation is made of acrylic with a thickness of 0.5 cm. The clear acrylic sheet allows direct observation of the movement mechanism inside the box. Unlike the other walls, which are permanently bonded, this acrylic wall can be removed to provide access to the interior of the measurement box.

The movement mechanism inside the box consists of a robotic arm mounted at the

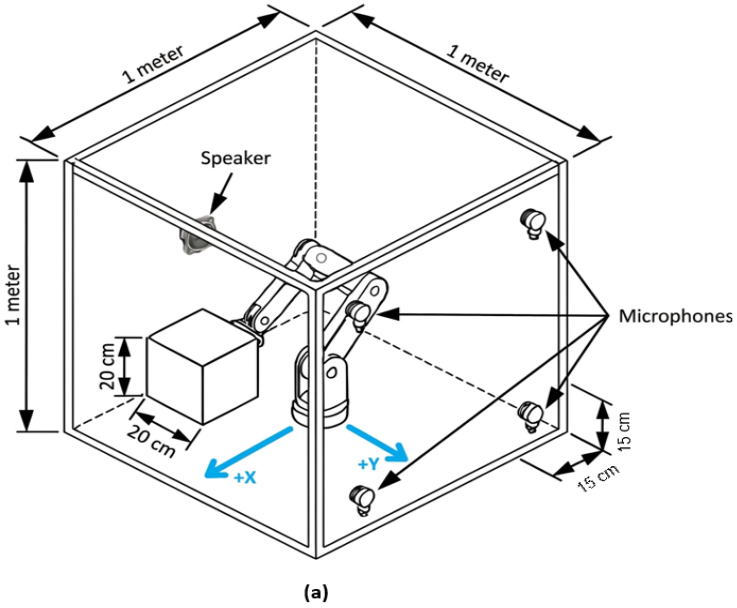
center of the floor. The arm has four Degrees of Freedom (DOF) and is used to control the position and orientation of a cube with a side length of 20 cm attached to its end. The cube is made of plastic, with walls 2 mm thick. For the measurements, a mono speaker is placed at the center of one wall to generate the acoustic stimulus inside the system. On the opposite wall, four capacitive microphones are positioned 15 cm away from each corner to capture the acoustic response of the measurement box. The measurement box was constructed inside a storage room and is shown in Fig. 4.1(b).

4.2. The Audio-Location Dataset for Model Training and Testing

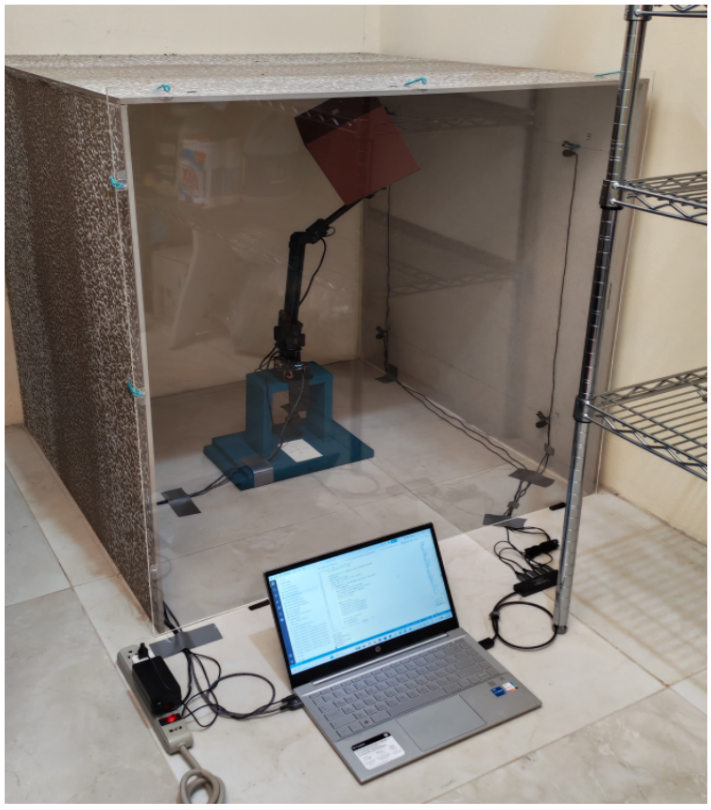
The dataset was collected over twenty-six recording sessions. In each session, the robotic arm maintained a fixed base angle and positioned the cube at specific coordinates and orientations. The initial orientation of the cube was set such that its sides were parallel to the measurement box walls. The variables modified during each measurement were the cube's location and orientation. For each location, nine different orientations were recorded. After completing these nine measurements, the robotic arm lowered the cube by decreasing its position along the Z axis, and the recording process was repeated for the new location. This procedure continued until the cube reached the bottom of the measurement box.

Audio from four channels was recorded. The acoustic signal was emitted four times, and the response was captured one microphone at a time. Each recording session consisted of 270 recordings, each containing four channels. A movement of 10 degrees in the base of the arm define the begining of a new recording session, with all the recording sessions producing a cylindrical envelope for the measurement area.

The coordinates of each measurement, obtained from the robotic arm, are shown in Fig. 4.2. Each blue column in the three-dimensional plot corresponds to one recording



(a)



(b)

Figure 4.1: Measure box: (a) Conceptual sketch for the acoustic measurement box, and (b) actual implementation for the data acquisition task.

session, and each point represents one of the nine orientations captured at that location. The total number of sessions was limited by the rotational constraint of the robotic arm's base, which allowed only 270 degrees of movement.

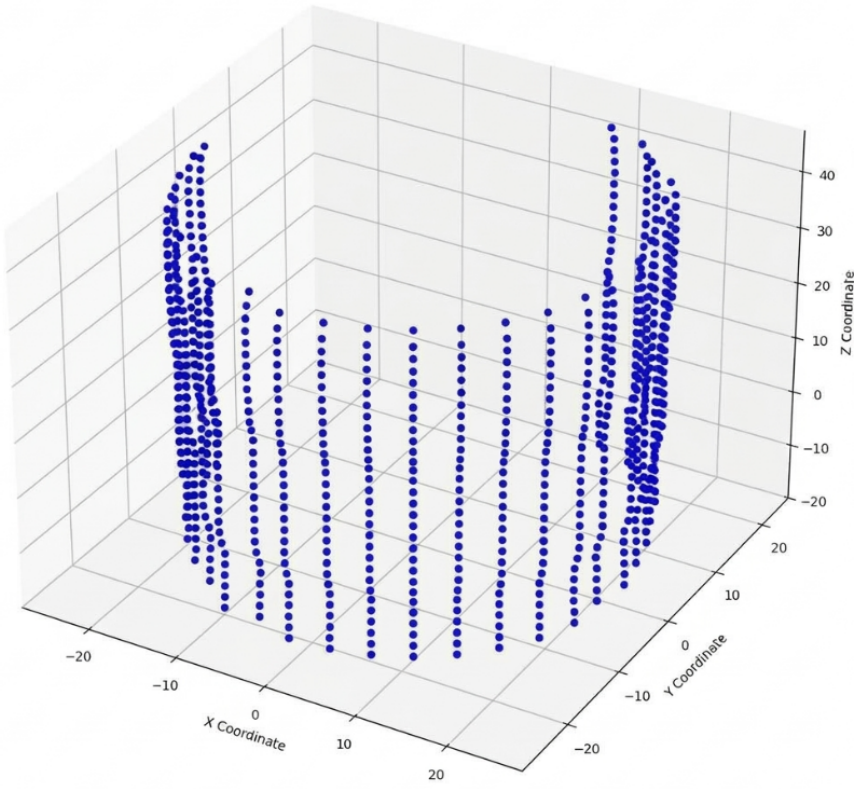


Figure 4.2: Scatter plot of cube locations for each recording in the dataset, the blue columns represent the 26 sessions, and each point in the column represents the nine orientations captured at that location.

The source audio signal was created by extracting an arbitrary 0.3-second segment from an acoustic melody. This type of audio was selected because of its suitability for acoustic environment recognition, as demonstrated by the melodies used in [33]. The signal's spectrum extends to 10 kHz, with higher-magnitude components concentrated at low frequencies. Before the source signal is played, the recording process begins by opening the capture window for one second. The signal is recorded four times, once for

each microphone. After these recordings are complete, the cube is moved to the next orientation. This process is repeated until all nine orientations at that location have been recorded, after which the cube is moved to a new location.

4.3. The Audio-Location Processing Pipeline

Figure 4.3 shows the processing pipeline for the acoustic signals. The microphone array in the measurement box records four signals, which form the four audio channels used as input to the pipelines. In the first box in the diagram AudAlign, the signals are aligned to compensate for variation in their start times caused by the operating system's process scheduler. Alignment is performed by exploiting the equal distances between the source and each microphone; therefore, the magnitude of the direct sound can serve as a reference. Also in AudAlign the signal integrity is validated. Each recording is divided into a main segment and a tail (the reverberation segment). The signal power is calculated from the main segment, and the tail duration is measured. Recordings whose signal power and tail duration fall within predefined limits are considered valid and included in the dataset. The resulting dataset is used in four scenarios: two use CNN-based pipelines, and two use ESN-based pipelines. The CNN-based pipelines require the time-series data to be converted into image representations, whereas the ESN-based pipelines use a transformation from audio to spectral sequences. In addition, beamforming is used to address acoustic superposition by dividing the room into four regions and representing the reverberation originating from each region in a separate set of channels. These channels are also provided as input to the CNN and ESN estimators.

4.4. Response Variables and evaluation metrics

The networks in this research are configured as regression models, the outputs are the predicted coordinates (X, Y, Z) and orientation of the cube in the room. The

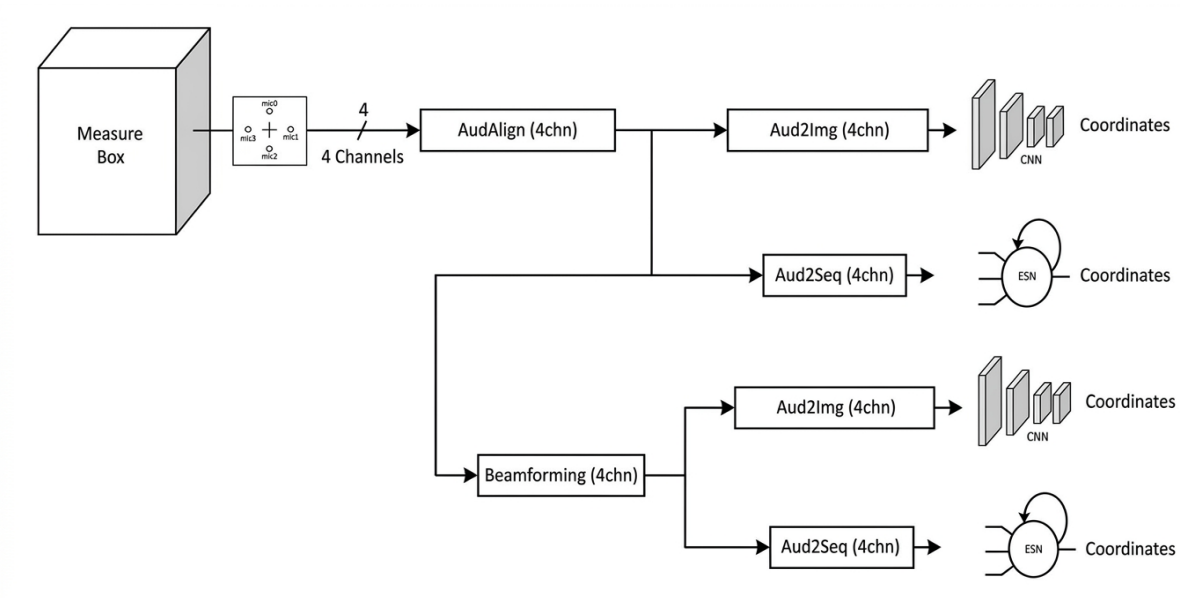


Figure 4.3: Acoustic to location processing pipeline

response variables are the error between the predicted and actual values of these outputs, which are calculated as mean of the absolute difference between the predicted $\hat{y}_i$ and actual values y_i as shows in equation 4.1. The Mean Absolute Error (MAE) is used as the performance metric for evaluating the models. The MAE is a widely used metric for regression tasks, as it provides a clear measure of the average magnitude of errors in predictions, without considering their direction. Lower MAE values indicate better model performance, as they reflect smaller discrepancies between predicted and actual values [34].

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{4.1}$$

Another way to evaluate a regression model is to measure the correlation between the actual and predicted values. The most widely used measure is Pearson's correlation coefficient, defined as follows:

$$r = \frac{\sum_{i=1}^{n}(y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2 \sum_{i=1}^{n}(\hat{y}_i - \bar{\hat{y}})^2}} \in [-1, 1], \tag{4.2}$$

where y_i and $\hat{y}_i$ are the actual and predicted values, respectively, and $\bar{y}$ and $\bar{\hat{y}}$ are their corresponding means. The numerator represents the covariance between the actual and predicted values, while the denominator normalizes this covariance by the product of the standard deviations of the actual and predicted values, ensuring that the correlation coefficient r lies within the range $[-1, 1]$.

This metric provides additional insight into the model's performance, complementing the MAE by indicating how well the predictions follow the trend of the actual values [34].

4.5. Data Recording for Model Evaluation

To evaluate the performance of the models, a five-fold cross-validation approach is used. In this approach, the dataset is divided into five subsets, and the model is trained and tested five times, each time using a different subset as the test set and the remaining subsets as the training set. This process allows for a more robust evaluation of the model's performance and helps to mitigate any potential bias that may arise from using a single train-test split. The total 7020 samples are divided into five folds, with each fold containing near to 1,404 samples (A 80:20 ratio). The performance of the model is evaluated considering the Mean Absolute Error (MAE). The structure of the five-fold cross-validation and the corresponding performance metrics are summarized in Table 4.1. Due to the factor and levels we have 4 model variation to consider, and for each model variation the coordinates X, Y, Z, and orientation are evaluated, resulting in a total of 20 MAE to obtain for evaluation.

Fold Iteration	Training Set	Test Set	Performance Metric
Fold 1	Fold 2, Fold 3, Fold 4, Fold 5	Fold 1	MAE_1
Fold 2	Fold 1, Fold 3, Fold 4, Fold 5	Fold 2	MAE_2
Fold 3	Fold 1, Fold 2, Fold 4, Fold 5	Fold 3	MAE_3
Fold 4	Fold 1, Fold 2, Fold 3, Fold 5	Fold 4	MAE_4
Fold 5	Fold 1, Fold 2, Fold 3, Fold 4	Fold 5	MAE_5

Table 4.1: Five-fold cross-validation structure and performance summary.

4.6. Tool Used for Models Comparison

In the literature, the use of parametric tests to compare machine learning models is often discouraged because the assumptions they require cannot always be taken for granted for learning algorithms [13]; nonparametric tests are therefore preferred. Two nonparametric approaches recommended for comparing regression models are the Wilcoxon signed-rank test and the Friedman test. The Wilcoxon signed-rank test estimates whether the differences between two models are statistically significant with respect to a specific evaluation metric. However, it is not recommended for comparing several models because doing so increases the risk of Type I errors. The Friedman test is used to compare more than two models [34].

Because performance metrics are obtained through cross-validation, the observations are not independent: the training sets overlap across folds. This dependence complicates the comparison of multiple models using the Friedman test and other traditional statistical tests [13].

On this problem, Bayesian approaches have been proposed as a more suitable alternative for comparing multiple models when the observations are not even independent, as they can properly account for the uncertainty and dependencies in the data [4; 9; 30]. These approaches provide a coherent probabilistic framework to model the rankings of the learing algorithms based on their performance metrics and make inferences about their relative performance.

In this work, we adopt the Bayesian Plackett-Luce (PL) framework described in

[9] to compare the four regression models base on the MAE as the main performance metrics obtained through cross-validation. This method allows us to properly account for the dependencies and uncertainties in the performance metrics obtained through five fold cross-validation. The procedure to compare multiple regression models using the Bayesian approach is implemented with the steps decribed in [9] and detailed as follows:

1) Create a matrix of performance metrics obtained through cross-validation for each regression model, where each row corresponds to a fold and each column corresponds to a model and from this matrix a ranking matrix is derived, where each element represents the rank of a model in a specific fold based on its performance metric.

2) Use the Plackett-Luce (PL) model as framework to model the probability distribution of the machine learning model rankings:

Parameter Interpretation: The parameters are a weight vector $\mathbf{w} = (w_1, \ldots, w_n)$ where $\sum_{i=1}^{n} w_i = 1$ and $w_i > 0$. Each weight w_i directly represents the marginal probability that algorithm i is ranked in first position (i.e., is the best algorithm).

Likelihood Formulation: Under the PL model, given the ranking for a fold k is $\sigma^{(k)} = (\sigma_1^{(k)}, \sigma_2^{(k)}, \sigma_3^{(k)}, \sigma_4^{(k)})$, the probability of observing this ranking is calculated as:

$$P(\sigma^{(k)} \mid \mathbf{w}) = \prod_{r=1}^{3} \frac{w_{\sigma_r^{(k)}}}{\sum_{m=r}^{4} w_{\sigma_m^{(k)}}} \tag{4.3}$$

3) A prior distribution is specified for the weight vector $\mathbf{w}$. A common choice is the Dirichlet prior:

$$P(\mathbf{w}) = \mathrm{Dir}(\mathbf{w}; \boldsymbol{\alpha})$$

where $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_n)$, with $\alpha_i > 0$ the hyperparameters of the prior, and set to $\alpha_i = 1$ for a non-informative prior.

4) Compute the posterior distribution of the weight vector $\mathbf{w}$ given the observed rankings. Using Bayes' theorem, the posterior is proportional to the likelihood times

the prior:

$$P(\mathbf{w}|R) \propto P(R|\mathbf{w}) \cdot P(\mathbf{w})$$

$$P(\mathbf{w}|R) \propto [\prod_{\sigma \in R} P_{PL}(\sigma|\mathbf{w})]^{\alpha} \cdot P(\mathbf{w}) \tag{4.4}$$

Where α is a deflation factor $(1+K\rho)^{-1}$ with K being the number of folds and ρ the correlation between training and test sets, defined by Nadeau and Bengio in [32] that adjusts the likelihood due to the fold cross-validation penalty applied in the Bayesian Plackett-Luce model as described in [21].

Because the posterior distribution presented in Equation (4.4) lacks a closed-form analytical expression, Markov Chain Monte Carlo (MCMC) sampling is employed to draw a representative set of weight vector samples from the posterior distribution $P(\mathbf{w}|R)$

5) Once the posterior samples of the weight vector $\mathbf{w}$ are obtained, counting the times the weight w_i is the largest among all weights allows to estimate the posterior probability that algorithm i is the best.

Chapter 5

Design

This chapter presents the models use to test the hipotesis. As the ground truth a CNN estimator is selected to explore the the capability of a model to learn and estimate from the acoustic signals the location of an object in a tridimentional space. The input trasformation to images and the changes requiered to the network to work as an estimator are mentioned. Also, in this section the design of a memory eficient network in the form of an ESN estimator is presented. A justification for the ESN model architecture selection, the input trasformation and training and evaluation process is provided. Finally, the beamforming mechanism intended to highlight acoustic echoes from specific regions is described.

5.1. CNN Estimator

The related work presented in seccion 2.6.2 provide insights respect the potential use of neural network to estimate the location of an object using acoustic echoes, and considering the especific findings in [35] is that a metric is extrapolated for the magnitude of estimation error. Due the fact that the dataset was new and unexplored, the model choose to establish a reference point was a CNN, this with the premise that a CNN have enogh representational capacity to learn from an especific acostic trasformation the function that maps the input features to espatial coordinates, providing valuable information if our original goals can be achieved with a more restricted network as is the case of an ESN.

5.1.1. Log-Mel spectrogram as images

To train a CNN on the dataset, a Log-Mel transformation is applied to the audio signals to generate spectrograms that can be used as network input. Each audio signal is 35,000 samples long. A short-time Fourier transform (STFT) is computed using a 2,048-sample window and a hop size of 512 samples, yielding 68 frames, as described

in Equation 5.1. An additional Mel filter bank compresses the spectral data into 128 bands, making it easier for machine-learning models to process [1].

$$\text{Frequency frames} = \frac{signal_length}{window_step} \tag{5.1}$$

Finally to reduce feature variance in the mel bands a logaritmic trasformation is perform. The resulting Log-Mel matrix is transform to images of 570x370 pixels per location-orientation as the one shown in Figure 5.1.

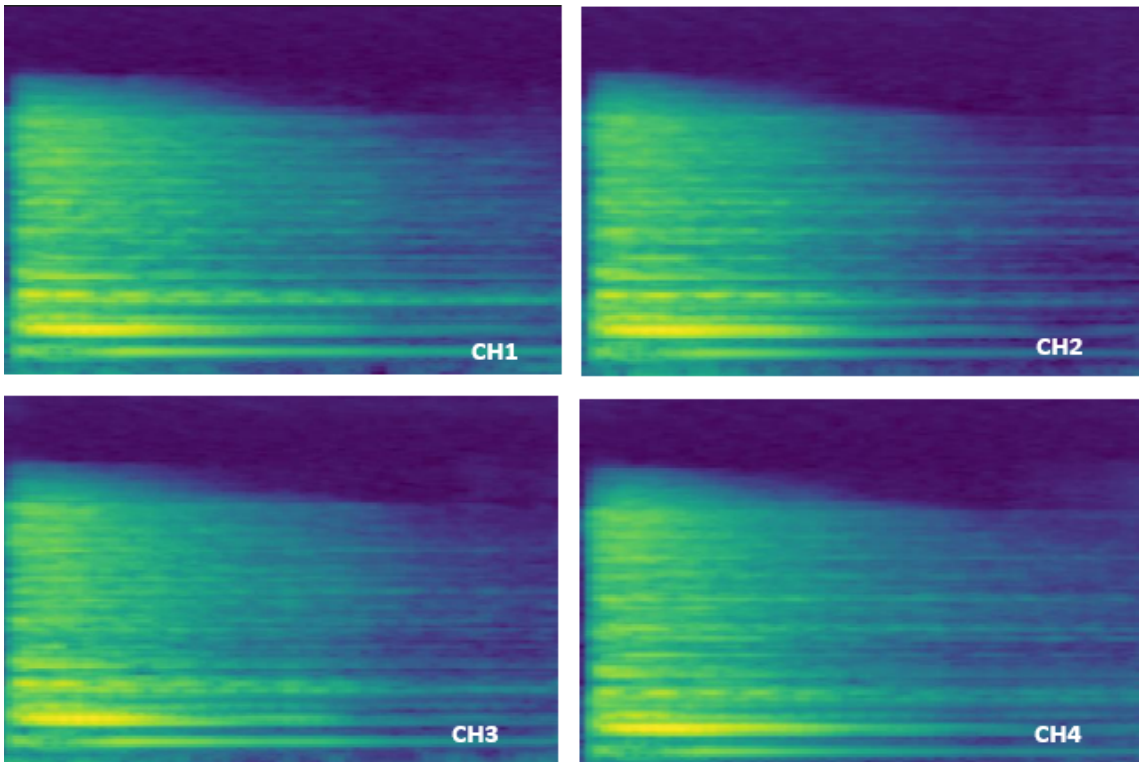


Figure 5.1: Log-Mel spectograms correspondig of the four channels of signals captured at session 1, location X=22.8cm, Y=-0.94cm, Z=-9.43cm and orientation 50 degrees.

5.1.2. CNN Architecture

The model architecture selected for the task of classifying and estimating the location–orientation of the cube inside the measure box was the ResNet-18 network with a total of 11,689,512 trainable parameters, chosen for its robustness, simplicity,

and suitability for inference in resource-constrained environments [5]. Two modifications were applied to the standard network structure. First, the input layer was extended to accept a stack of four RGB images, one corresponding to each audio channel. Second, for the regression task, the softmax output layer was replaced with a single output feature using a sigmoid activation function.

5.1.3. CNN Training and Evaluation

The network was trained to minimize the mean squared error (MSE) using the adaptive moment estimation (Adam) optimizer. Five-fold cross-validation was used: in each fold, four folds were used for training and the remaining fold for testing, corresponding to an 80:20 train-test split. The model was trained for 40 epochs in each fold, with 350 batches of 32 samples per epoch for a total time of training per fold near to 1.2 hours.

Before training, the target values for X, Y, and Z were normalized. Min-max normalization scaled each value to the range $[0, 1]$ to match the network's output range. An additional normalization step was applied to X and Y because the points lie along a circular trajectory. The robotic arm's rotational movement in constant 10-degree increments means that the projections of these coordinates are expressed as multiples of sine and cosine functions. Consequently, the coordinate values are not uniformly distributed. The arcsine transformation shown in Equation 5.2 was therefore applied to obtain an equidistant mapping.

$$\text{label_arcnorm} = \frac{\arcsin(2\text{label_norm} - 1) + \frac{\pi}{2}}{\pi} \tag{5.2}$$

After training, the model was switched to evaluation mode, and the test set was processed to obtain the estimator's predictions. The inverse of the transformation in Equation 5.2 was applied to denormalize the predictions. Estimation errors were then calculated as the differences between the predicted and ground-truth coordinates. The

mean absolute error (MAE), error distribution, and percentage of predictions within each error range were calculated from the test results.

5.2. ESN Estimator

Echo State Networks (ESNs) are well suited to sequential data because their rich temporal memory and nonlinear dynamics enable them to learn complex patterns, as in recurrent neural networks (RNNs). Unlike conventional RNNs, however, ESNs do not require training the recurrent part of the network. They also tend to be compact, making them a memory-efficient choice for the estimator proposed in this thesis.

5.2.1. Mel spectrogram as sequence

For the ESN, each audio channel is cropped to 20,000 samples to remove the low-energy tail of the signal. The cropped signals are then transformed into sequences of Mel-spectrum values using a process similar to that described in the CNN subsection on Log-Mel spectrograms. To reduce computation time, the short-time Fourier transform (STFT) uses a window length of 1,024 samples and a hop size of 512 samples, yielding 40 frames. A Mel filter bank compresses the data into 128 bands, a resolution adequate for acoustic scene and direction-of-arrival identification [2]. The resulting bands are converted to decibels, producing a sequence of 40 time steps with 128 features per channel. Finally, min-max normalization is applied to the input features to mitigate issues caused by large differences in magnitude.

5.2.2. ESN Architecture

The development of the ESN architecture begins with the vanilla flavor of the ESN all set to work as an clasifier. for exploration and evaluation the network is first configure as an clasifier due that this task is easier than regression [45], so the earlier evaluations

and changes on the ESN are conducted in it clasification mode.

The base ESN had as input four mel spectograms sequences of 128 bands each, for a total of 512 inputs, the neurons use tanh activation function for reservoir and the feedforward network. The output softmax layer clasifiy into 26 classes based on the session in which the audio is recorded.

The parameters for the networks are show in the table 5.1:

Table 5.1: Key parameters of an Echo State Network (ESN)

Parameter	Description
`reservoir size`	The total number of internal recurrent neurons within the reservoir, determining the dimensional capacity to represent complex dynamical states.
`spectral radius`	The largest absolute eigenvalue of the recurrent weight matrix, governing stability, memory decay rate, and the Echo State Property (ESP).
`sparsity`	The fraction of zero-weight connections within the reservoir matrix, decoupling internal sub-dynamics and lowering computation costs per update.
`leaky rate`	The leaking integration parameter $[0, 1]$, controlling how fast the reseirvoir forget previous states.

The spectral radius that control how memory decay in the reservoir is desire to have it as hier as posible in order that the network present a long term memory behaviour, the maximun value recommended had a boundary of 1 to ensure the ESP to warranty that the internal states of the reservoir will converge over time avoiding instability. For the ESN presented in this work a value of 0.99 was choose as spectral radius for the reservoir matrix. The leaky rate also controlls how fast the reservoir forget and it set to 0.1 to avoid memory loss and keep as much as posible of the history of the input sequence in the reservoir.

Sparsity was set to 0.1 to maintain sufficient connectivity among the reservoir neurons. Because reservoir size affects both memory use during training and the number of trainable weights in the feedforward network, a search was conducted to identify a favorable balance between accuracy and model size. Figure 5.2 shows the

relationship between accuracy and reservoir size. Accuracy increases with reservoir size up to 1,500 neurons; beyond that point, further increases do not necessarily improve network performance.

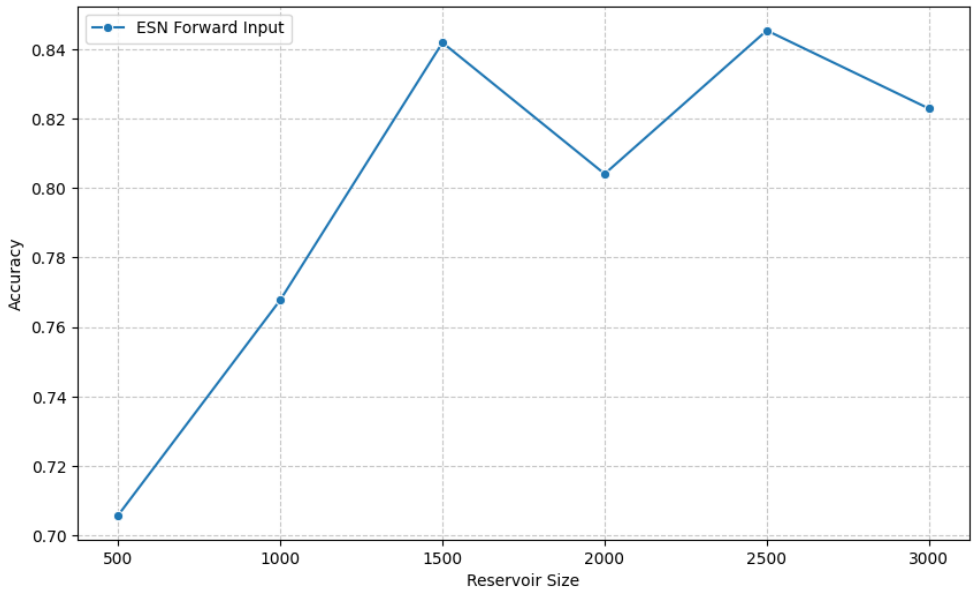


Figure 5.2: Network accuracy vs. reservoir size for 50 epochs runs.

A test of the described configuration achieved less than 95% accuracy after 100 training epochs. To improve performance, the number of features extracted from the reservoir was increased by capturing its state at multiple points in the sequence. Four configurations were evaluated: one capture at the end of the sequence; two captures, at the midpoint and end; three captures, at 33%, 66%, and 100% of the sequence; and four captures, at 25%, 50%, 75%, and 100%. The results are presented in Table 5.2. The four-capture configuration achieved the highest mean accuracy and was therefore selected to provide features to the feedforward network.

The table 5.3 summarize the final values selected for the clasifier, with those values an acuracy of 98.42% was achieved on 150 epoch training.

The classification results demonstrated that the ESN could learn patterns from the input sequences to distinguish the sessions in which the recordings were made. This

Table 5.2: Performance Across Intermediate Steps

Intermediate Steps	Run1	Run2	Run3	Mean
1 step	0.7692	0.7728	0.8262	0.7894
2 steps	0.8946	0.8291	0.8340	0.8525
3 steps	0.8547	0.8889	0.9024	0.8820
4 steps	0.9231	0.8996	0.9160	0.9129

Table 5.3: Echo State Network (ESN) Hyperparameters

Parameter	Value
Reservoir Size	1500
Spectral Radius	0.99
Sparsity	0.1
Leaky Rate	0.1
Intermediate Captures	4

classifier served as the basis for the regression network. The output layer was changed from 26 units to one, and two additional hyperparameters were tuned using grid search: (1) the reservoir sparsity and (2) the hidden-layer size. To improve the ESN's learning capacity, the single-layer feedforward network was replaced with a two-layer network containing a hidden layer whose number of units is specified by the hidden-layer size. The search results for these parameters are shown in Figure 5.3. To balance model performance and size, a sparsity of 0.7 and a hidden-layer size of 20 were selected for the ESN estimator.

Table 5.4: Error Distribution Comparison Between Standard ESN and MultiHead ESN

Error Range (cm)	ESN Count (%)	MultiHead ESN Count (%)
$(-1, 1)$	39.80	76.65
$(-2, 2)$	62.31	95.01
$(-3, 3)$	76.72	98.38
$(-4, 4)$	85.16	99.09

After 1,200 training epochs, the ESN estimator achieved a mean absolute error (MAE) of approximately 2 cm. The error-range percentages are reported in Table 5.4.

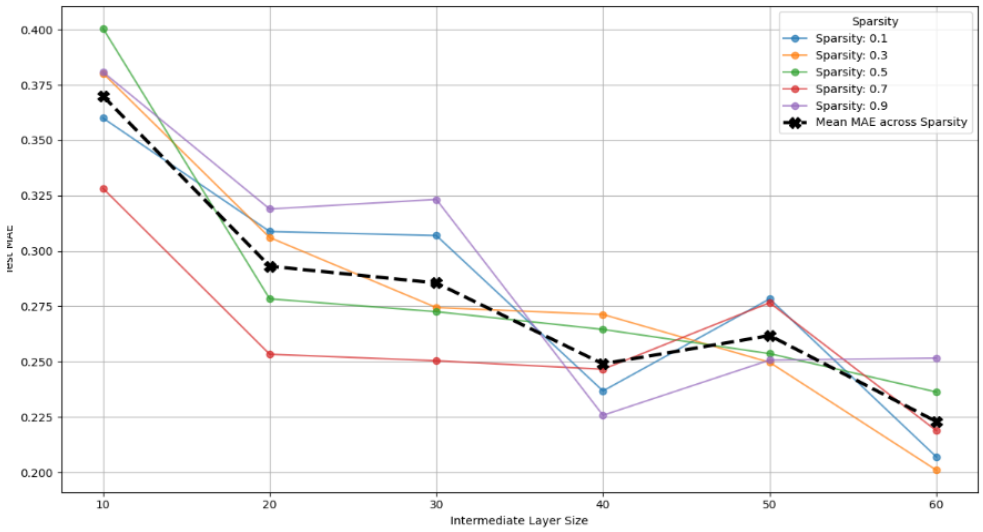


Figure 5.3: Test MAE as a function of Intermediate Hidden Layer Size for different Sparsity values

To further reduce the MAE, the architecture was extended with a multihead feedforward network. This change was motivated by the ESN's performance as a classifier: it could distinguish recording sessions, including sessions with nearby positions. However, in the regression task, the estimation errors were sometimes larger than the distance between adjacent sessions, suggesting that the network had difficulty learning the mapping from reservoir features to output coordinates.

The classifier used a separate output for each class, allowing it to learn a distinct mapping for each class. Inspired by this approach, the multihead network divides coordinate prediction into separate tasks: one head predicts positive coordinate values, another predicts negative values, and a third predicts the coordinate sign. The sign-prediction head selects which of the positive or negative value heads provides the coordinate estimate. This approach substantially improved the ESN estimator's performance, but occasional sign-prediction errors still caused large estimation errors. To address this issue, a fourth head was added to estimate the coordinate across its full range. Although this head is less accurate on its own, its prediction serves as a reference for the other heads. If its prediction differs from the selected positive or

negative value prediction by more than a threshold (a difference exceeds 100 cm, indicating a sign-classification outlier error), the model treats this as a possible sign-prediction error and selects the estimate from the opposite-sign head instead.The hybrid decision logic is shown in Algorithm 1.

For the sign and full range heads, the feedforward networks contains 60,000 trainable parameters each. The positive and negative coordinate heads that requiere to be more acurate have 120,000 trainable parameters each. This results in a total of 360,000 trainable parameters for the multihead feedforward network what is only near to 3% of the ResNet-18 CNN parameters. The multihead feedforward network architecture is shown in Figure 5.4. Its results, including an MAE of approximately 0.7 cm and improved error-range percentages, are reported in Table 5.4.

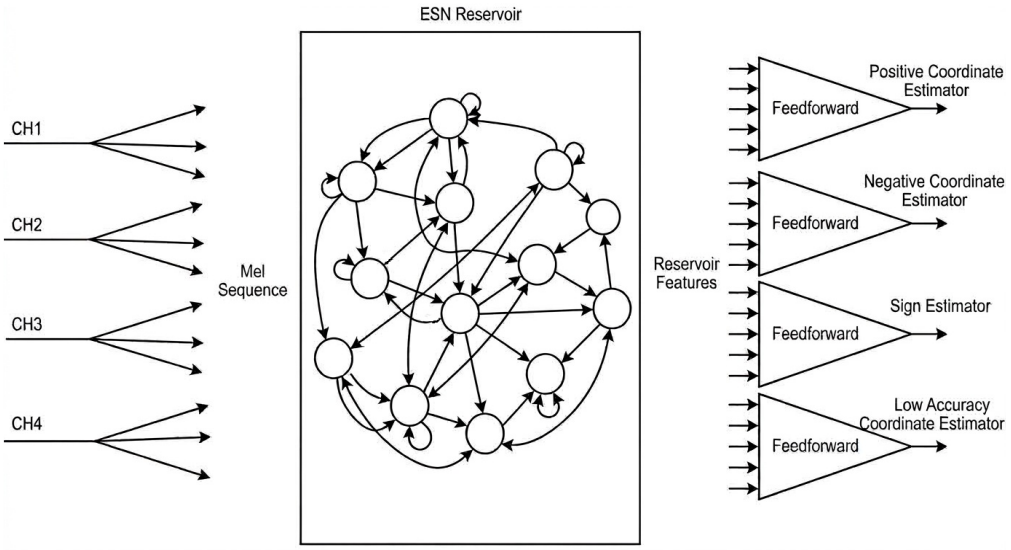


Figure 5.4: Multihead Network architecture for the ESN Estimator.

5.2.3. Multi-Head ESN Training and Evaluation

Each feedforward head was trained independently using a batch size of 32 and a learning rate of 5×10^{-5}. The positive and negative coordinate heads were trained

only on samples with the corresponding sign, whereas the overall-coordinate and sign-prediction heads were trained on all samples. The positive, negative, and overall-coordinate heads were trained with the Adam optimizer for 400 epochs. The sign-prediction head was trained for 800 epochs using binary cross-entropy with logits loss. The total training time for each fold was approximately 1.5 hours.

The models were trained and evaluated using five-fold cross-validation: four folds were used for training and one for testing in each iteration, corresponding to an 80:20 split. The target values for X, Y, and Z were normalized as described in Section 5.1.3. The network outputs were then denormalized to obtain the final coordinate estimates.

Algorithm 1 Hybrid Decision Logic for Coordinate Estimation with Outlier Mitigation

Require: Feature vector $\mathbf{x}$, Sign Head $\hat{s}(\cdot)$, Positive Head $\hat{y}_+(\cdot)$, Negative Head $\hat{y}_-(\cdot)$, Overall Estimator $\hat{y}_{\text{overall}}(\cdot)$, Outlier Threshold $\tau = 100$ cm
Ensure: Final Coordinate Estimation $\hat{y}$

// Phase 1: Sign Classification
pred_sign $\leftarrow \hat{s}(\mathbf{x})$ ▷ Predicts whether target is positive or negative
// Phase 2: Conditional Path Selection
if pred_sign ≥ 0 **then** ▷ Positive condition branch
    $\hat{y}_{\text{cand}} \leftarrow \hat{y}_+(\mathbf{x})$ ▷ Candidate conditional path prediction
    $\hat{y}_{\text{alt}} \leftarrow \hat{y}_-(\mathbf{x})$ ▷ Alternate path for fallback
else ▷ Negative condition branch
    $\hat{y}_{\text{cand}} \leftarrow \hat{y}_-(\mathbf{x})$
    $\hat{y}_{\text{alt}} \leftarrow \hat{y}_+(\mathbf{x})$
// Phase 3: Outlier Mitigation and Fallback Logic
$\hat{y}_{\text{ref}} \leftarrow \hat{y}_{\text{overall}}(\mathbf{x})$ ▷ Evaluate baseline overall estimator
$\Delta \leftarrow |\hat{y}_{\text{ref}} - \hat{y}_{\text{cand}}|$ ▷ Compute absolute discrepancy
if $\Delta > \tau$ **then** ▷ Potential critical sign classification failure detected
    $\hat{y} \leftarrow \hat{y}_{\text{alt}}$ ▷ Fallback to alternate estimator output
else
    $\hat{y} \leftarrow \hat{y}_{\text{cand}}$ ▷ Retain candidate conditional path prediction
return $\hat{y}$

5.3. Beamforming Implementation

In order to implement beamforming using the audio channels, the principal spectral components of the source signal used as the stimulus within the measurement box must be analyzed. For this purpose the top 10 frequencies with the highest magnitude are presented in Table 5.5, from which the most relevant frequency components can be identified, and the corresponding frequency response in the audio signal is illustrated in Figure 5.5 for the range of 0 Hz to 800 Hz.

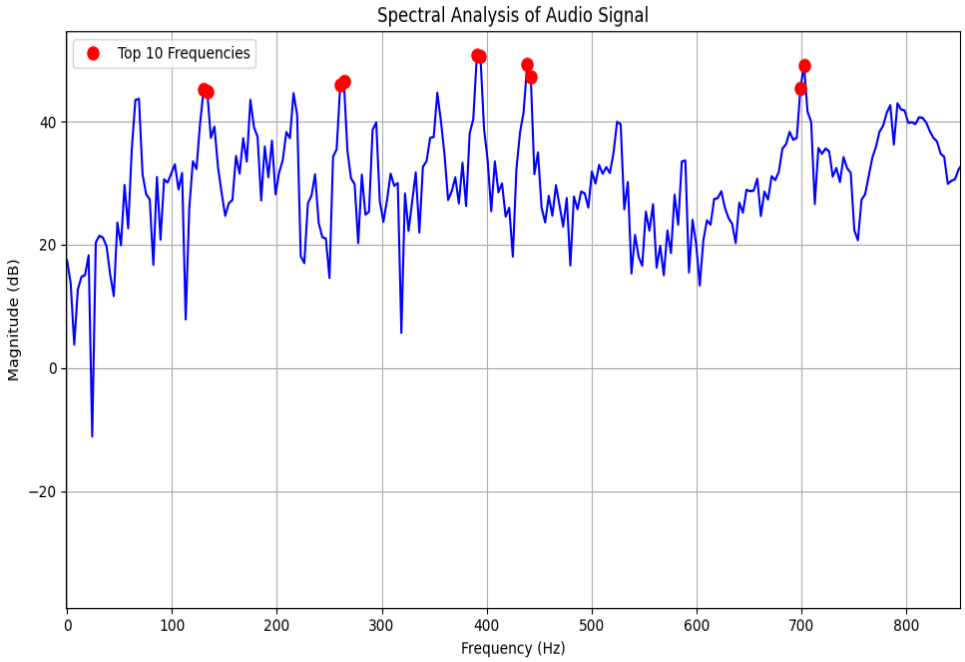


Figure 5.5: Frequency response of the stimulus signal showing the top 10 spectral components with the highest magnitude.

The microphones are arranged in pairs in a plane parallel to the X-Z plane, enabling binaural recording. Measurements along the X and Z axes can therefore be used to evaluate the effect of beamforming on position estimation. Figure 5.6 shows the midpoints of the sides perpendicular to the X-axis, which are treated as potential source locations for the beamforming procedure. To model sound arriving at both microphones from these locations or nearby regions, the difference between the two

Rank	Frequency (Hz)	Magnitude (dB)
1	390.72	50.71
2	394.14	50.58
3	438.70	49.31
4	702.61	49.17
5	442.13	47.26
6	263.91	46.60
7	260.48	45.96
8	699.18	45.32
9	130.24	45.19
10	133.67	44.88

Table 5.5: Top 10 frequency components of the stimulus signal with the highest magnitude.

acoustic path lengths must be calculated. Using trigonometry, the distance from point E to point G is 0.522 m, while the distance from E to point H is 0.9862 m. The difference, 0.4642 m, is the additional distance the sound wave travels to reach point H. Assuming a speed of sound of 346 m/s—the approximate value at 26 °C [46], which is close to the average temperature at the testing site—the time difference of arrival (TDOA) between the microphones is 1.34 ms. As shown in Equation 5.3, this value serves as the reference delay for beamforming. Applying this delay to one microphone signal aligns the signals for constructive interference and maximizes the coherent summation of the desired sound wave. According to Equation 2, given a microphone separation of 0.7 m, the minimum frequency for beamforming is 123.57 Hz. Figure 5.5 and Table 5.5 show that the stimulus contains its main frequency components above this threshold, making beamforming feasible in this setup.

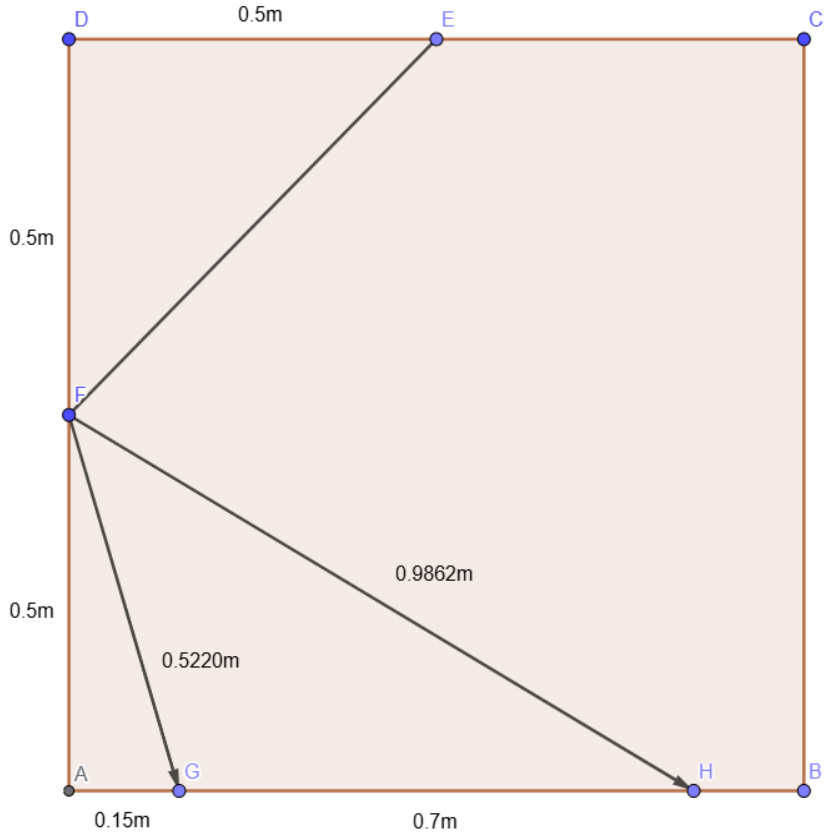


Figure 5.6: Geometric layout of the measurement box illustrating the acoustic path from the speaker located at E to the microphones positioned at G and H.

$$\Delta t_{\text{arrival}} = \frac{0.4642\,\text{m}}{346\,\text{m/s}} = 1.34\,\text{ms} \tag{5.3}$$

As an example, selecting the highest-magnitude components around 400 Hz for beamforming, combined with the computed time difference of arrival, enables an analysis of how the signal's main spectral components behave over the travel time Δt. At 400 Hz, the signal period is 2.5 ms. For a propagation delay $\Delta t = 1.34$ ms (as illustrated in Figure 5.7), the acoustic wave arrives at point G in an inverted-phase condition relative to point H, causing destructive cancellation when the two signals are summed. Consequently, introducing a delay equal 1.34 ms to the signal at point G

produces an alignemet of the signals resulting in constructive interference for signal when added to the signal from point H at 400 Hz. Furthermore, as the reverberant source shifts away from this target location, destructive attenuation penalizes the gain at these frequencies, fulfilling the objective of the beamforming operation.

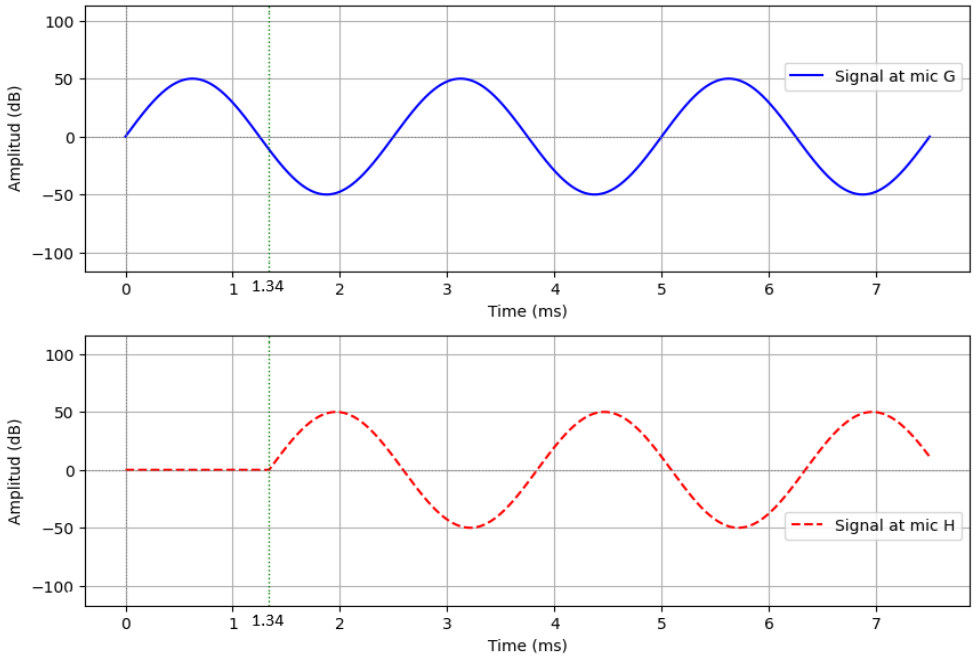


Figure 5.7: Acoustic waveforms at 400 Hz captured by the microphones located at points G and H.

Chapter 6

Results

This chapter presents the results obtained from the evaluation of the proposed models for acoustic inferencing tasks. First, the models are evaluated on their ability to classify the acoustic location of the cube inside the measurement enviroment. Then, a more broad evaluation is conducted where the models are assessed on their ability to do regression and properly estimate the location of the object inside the measurement box. For the regresion results, the figures of merit are presented for each model on each spatial dimension they are traing to predict. Finally a probabilistic evaluation is conducted to compare the performance of the models against each other.

6.1. CNN for Acoustic Classification

The first model trained and evaluated is the Resnet18 CNN, this network is powerfull for image classification tasks [5] and it used as the reference model for the acoustic inferencing tasks presented in this work.

6.1.1. CNN Model for Acoustic to Spatial Location Classification

The CNN was trained on a classification task where it had to classify the acoustic location of the sound source from the 26 possible locations, each one corresponding to a specific recording session characterized by the same X, and Y location and only changing the Z location. The network was trained on 20 epochs. After the training the network shows a accuracy of 99.4% on training and 98.59% on the test set. The confusion matrix is shown in Figure 6.1.

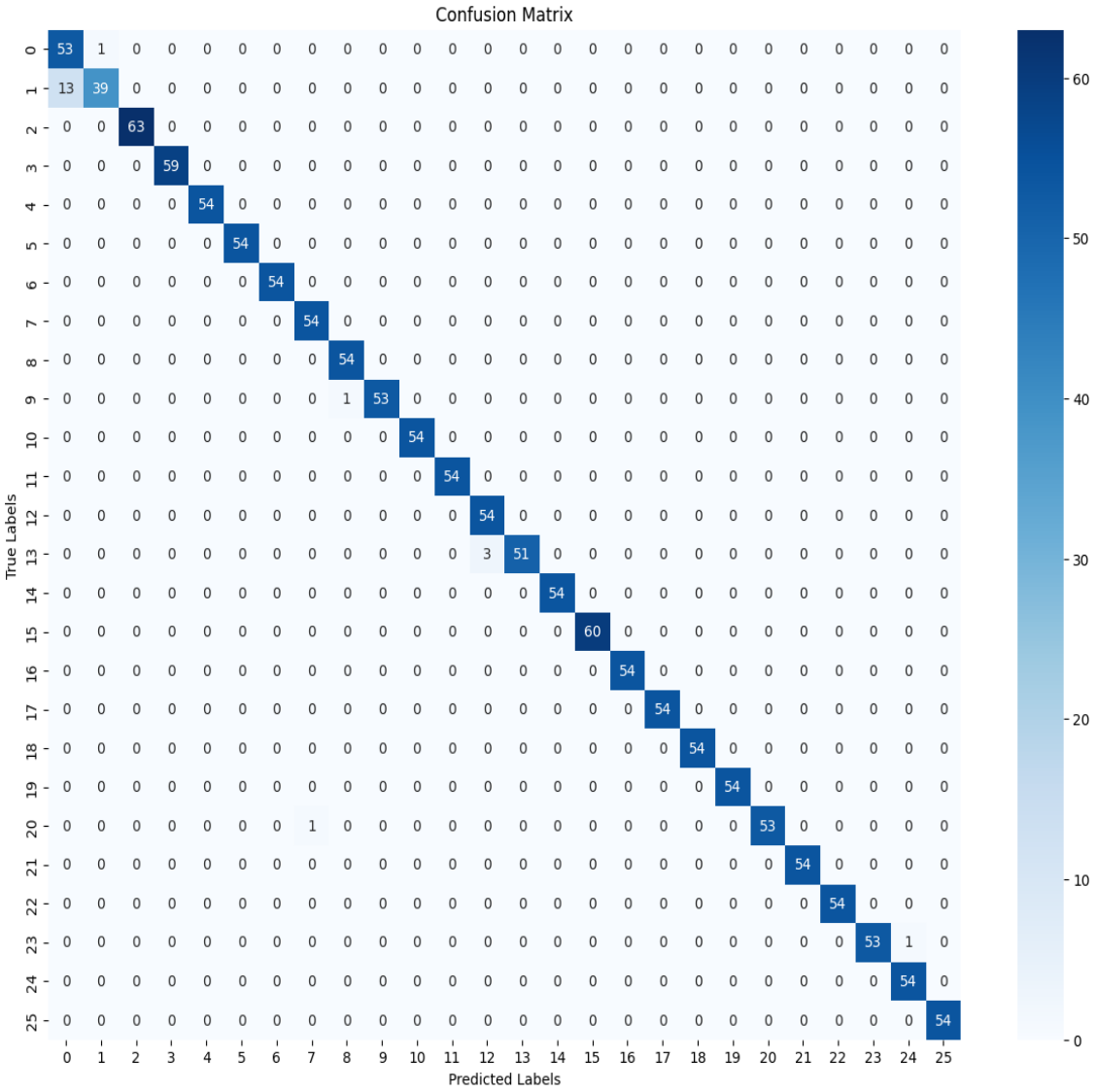


Figure 6.1: Confusion matrix for the evaluation of the CNN for the acoustic location classification task.

6.1.2. CNN Model for Acoustic to Orientation Classification

The second evaluation tested the CNN's ability to classify the cube's orientation among nine possible orientations. For each orientation, the cube was rotated by 10 degrees about one of its centroidal axes. The network was trained for 20 epochs and achieved an accuracy of 99.0% on the training set and 29.06% on the test set. The confusion matrix is shown in Figure 6.2; each class represents a cube orientation that

differs from the others by 10 degrees.

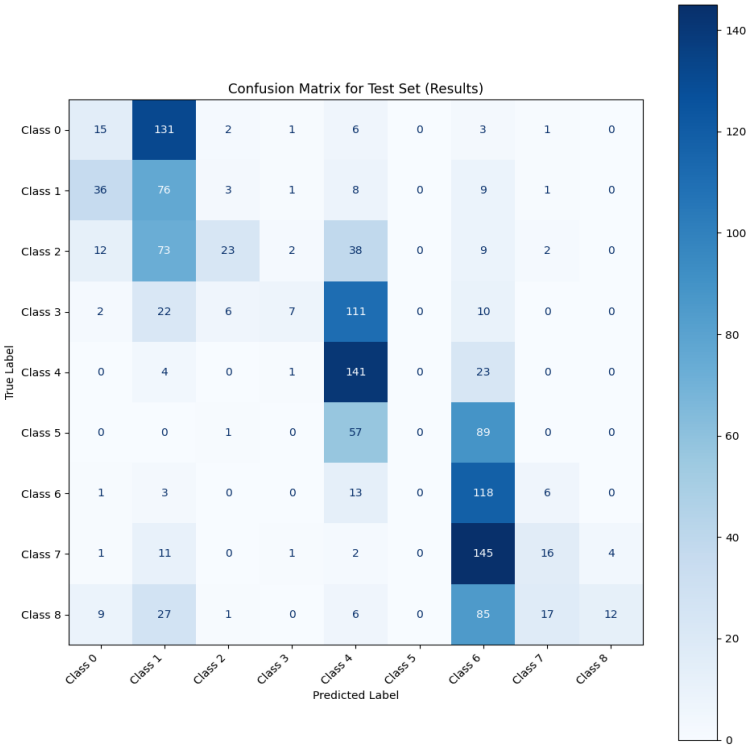


Figure 6.2: Confusion matrix for CNN acoustic orientation classification on test set.

Because the CNN performed poorly on the overall orientation classification task, its performance was further evaluated under more constrained conditions. As shown in Figure 6.4, three regions were defined to limit variation in orientation about the other Euler axes. Accuracy results for each region are presented in Table 6.1, and the corresponding confusion matrices are shown in Figure 6.3.

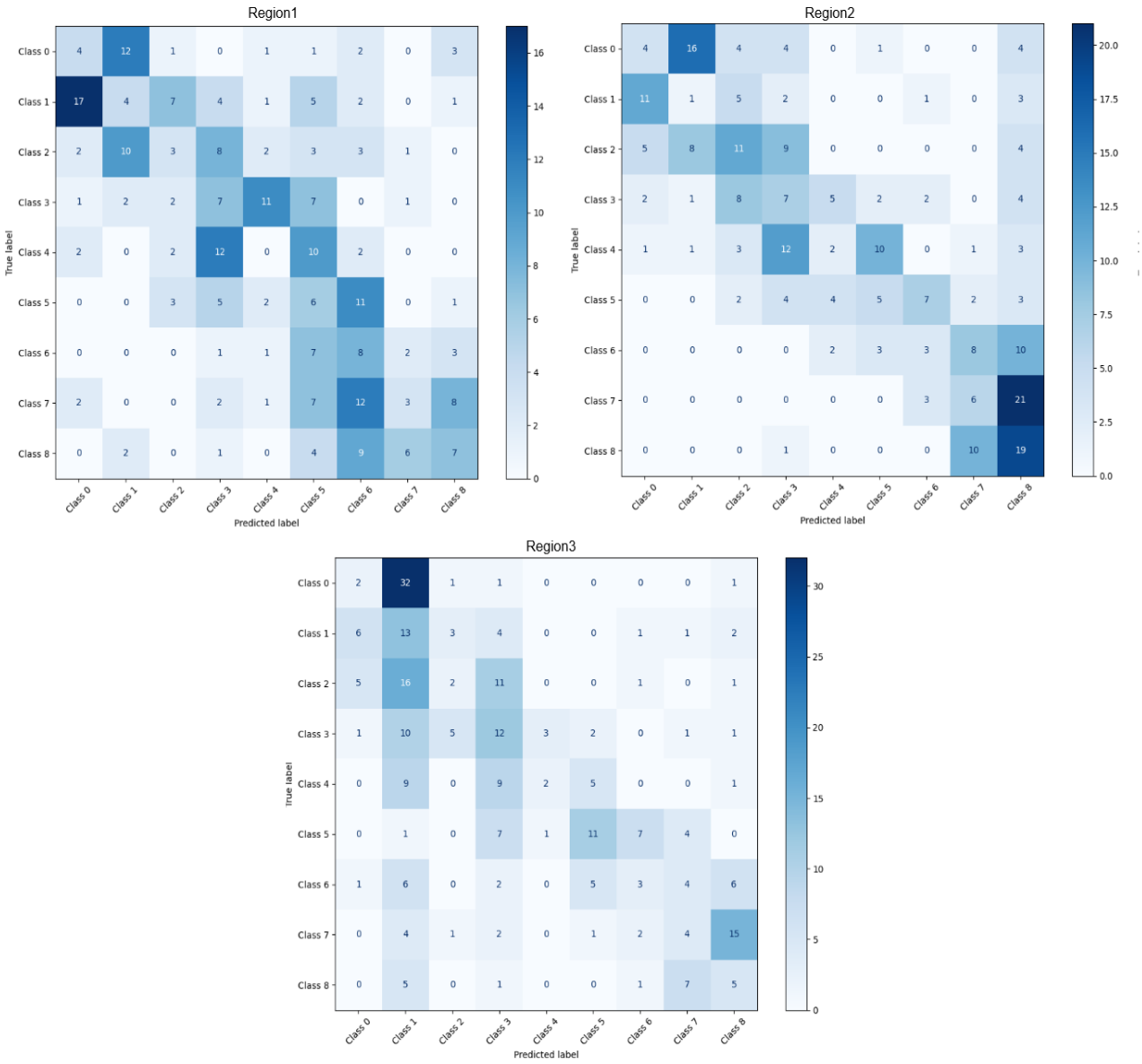


Figure 6.3: Confusion matrix for the three regions defined for orientation classification.

Table 6.1: Model Performance Across The Three Regions

Region	Training Accuracy (%)	Test Accuracy (%)
Region1	100.0	15.56
Region2	99.9	21.48
Region3	100.0	20.00

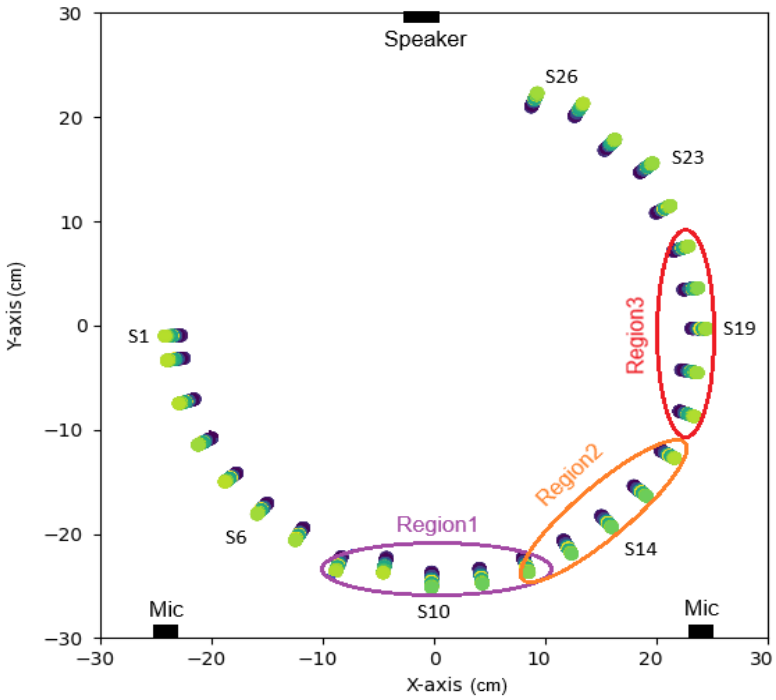


Figure 6.4: Session point used for cross-correlation analysis. the image also shows the three regions for analysis of the orientation classification task.

Given the even poorly performing test accuracies across the three regions, it is evident that the model struggles to generalize well beyond the training data. To investigate the similarity of the acoustic signal when the cube is rotated or the cube is moved, a cross-correlation analysis was performed for the center point of each sesion on each region, as illustrated in figure 6.4. The results of the cross-correlation analysis are presented in figure 6.5, showing the degree of similarity for different distances and orientations. This analysis helps to understand the limitations of the model in capturing the spatial variations of the acoustic signal.

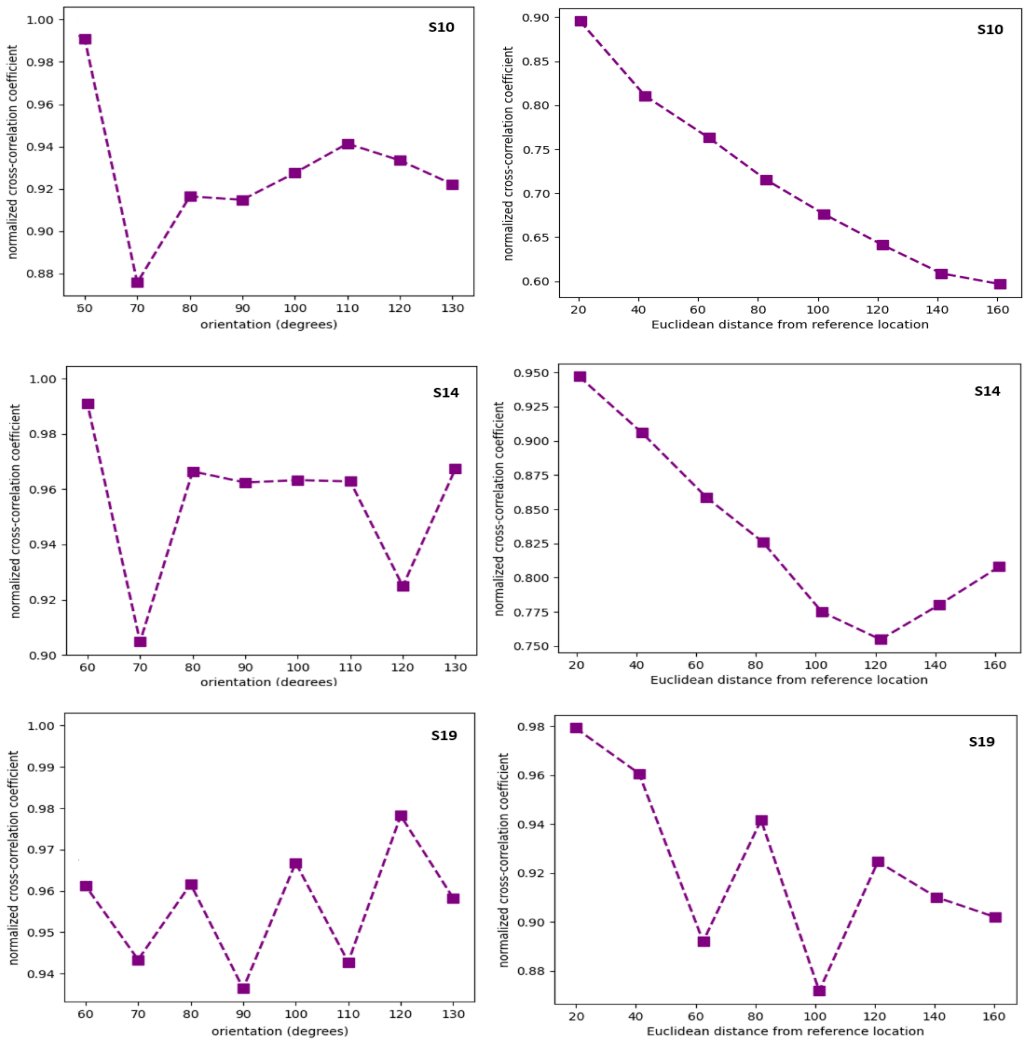


Figure 6.5: Cross-correlation results for distance and orientation for similarity analysis of the acoustic signal for the regions defined in the figure 6.4.

6.2. CNN for Acoustic to Spatial Location Regression

The CNN was evaluated as a regression model to estimate the cube's X, Y, and Z coordinates within the measurement box. It was trained for 40 epochs in each fold of a five-fold cross-validation scheme, with training taking approximately 1.5 hours per fold. The estimation-error histograms for the three coordinates are shown in Figures 6.6, 6.7, and 6.8.

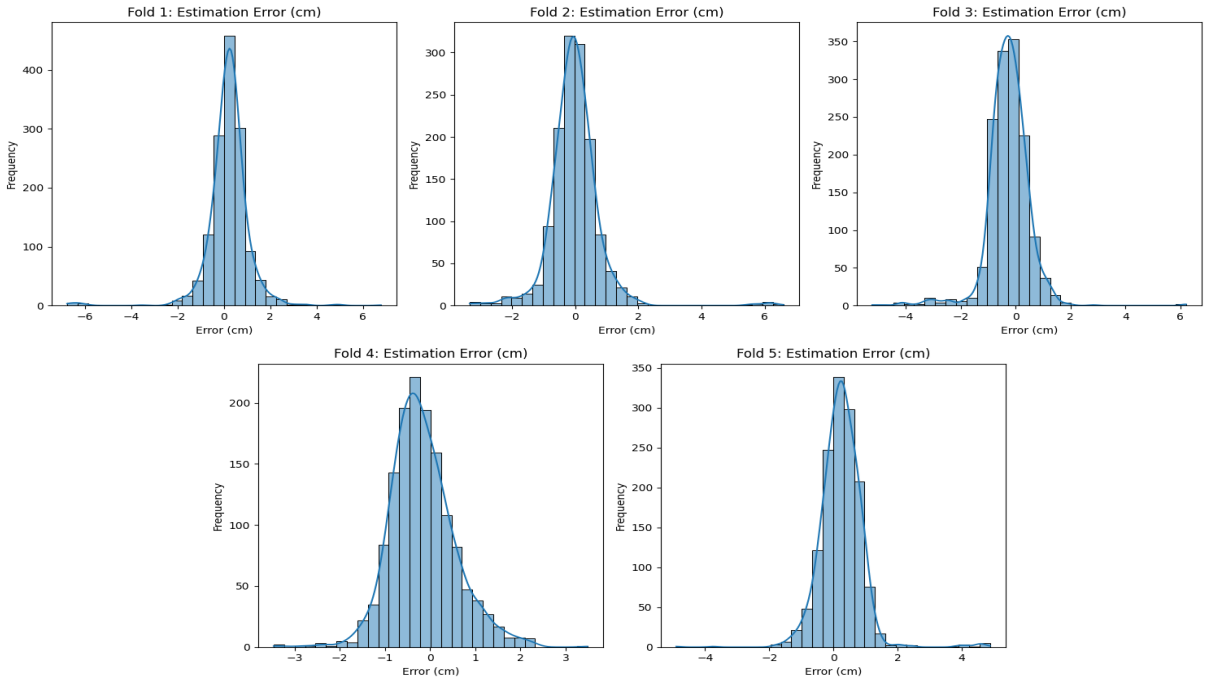


Figure 6.6: Histograms of estimation errors with the CNN for X coordinates across five folds.

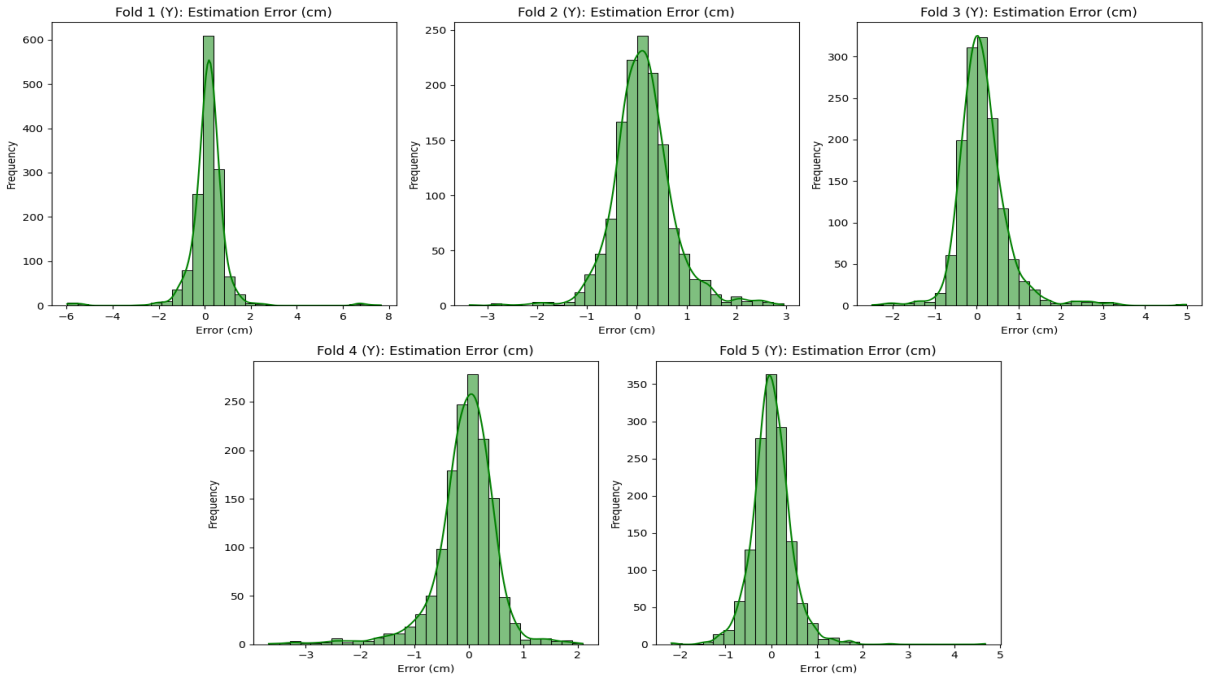


Figure 6.7: Histograms of estimation errors with the CNN for Y coordinates across five folds.

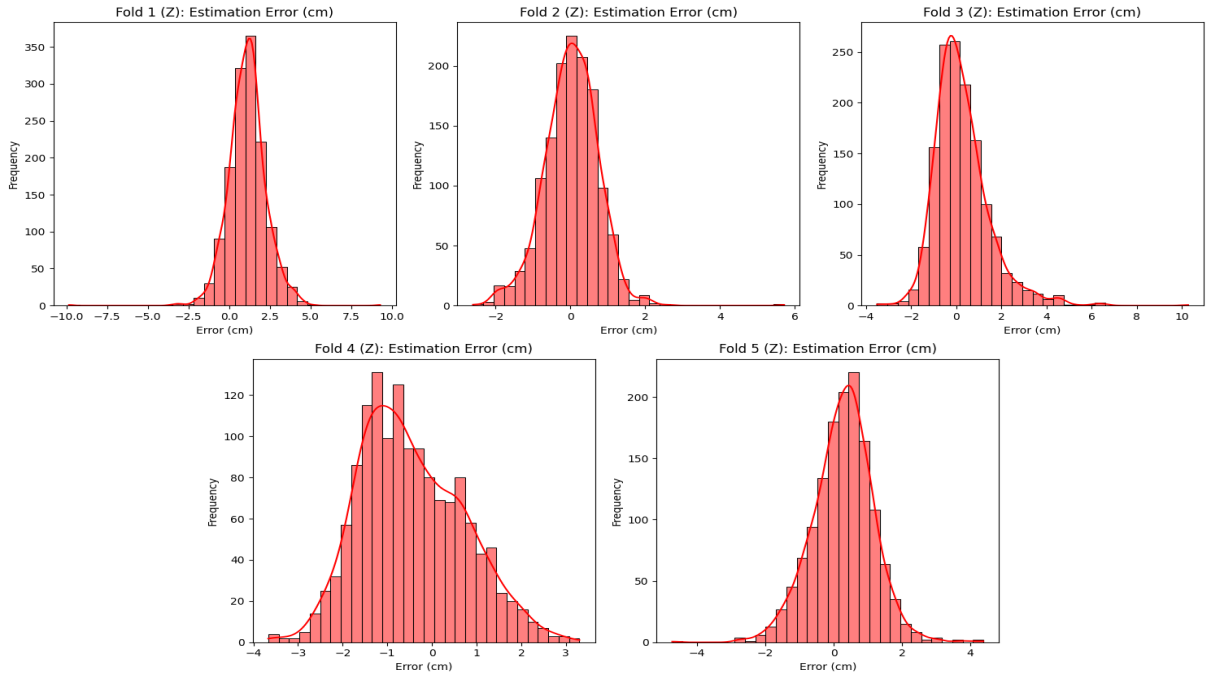


Figure 6.8: Histograms of estimation errors with the CNN for Z coordinates across five folds.

The main figure of merit for a regression model is the Mean Absolute Error (MAE), which measures the average magnitude of the prediction errors regardless of their direction. The MAE values for the three coordinates across the five folds are presented in Table 6.2.

Table 6.2: Mean Absolute Error (MAE) Across Folds for X, Y, and Z Coordinates for the CNN Regression Model

Folds	**MAE X (cm)**	**MAE Y (cm)**	**MAE Z (cm)**
Fold 1	0.6019	0.5029	1.3013
Fold 2	0.5231	0.4467	0.5549
Fold 3	0.5569	0.4069	0.8829
Fold 4	0.5764	0.3836	1.0473
Fold 5	0.5062	0.3144	0.7316
Mean ± SD	**0.5529 ± 0.0384**	**0.4109 ± 0.0706**	**0.9036 ± 0.2847**

A second key metric is the Pearson correlation coefficient, which measures the strength and direction of the linear relationship between two continuous variables. It

indicates how closely the data points cluster around a straight line. To understand how well the predictions of the regression model correlate with the ground truth, the Pearson correlation coefficients for each coordinate are shown in Table 6.3. In the evaluation of predictions against ground truth, a coefficient greater than 0.99 indicates that the model predictions are almost perfectly aligned with the actual targets.

Table 6.3: Pearson Correlation Coefficients between Ground Truths and Predictions for the CNN Regression Model

Folds	X	Y	Z
Fold 1	0.9970	0.9961	0.9979
Fold 2	0.9980	0.9985	0.9991
Fold 3	0.9969	0.9978	0.9982
Fold 4	0.9985	0.9977	0.9986
Fold 5	0.9978	0.9987	0.9987
Mean ± SD	**0.9976 ± 0.0007**	**0.9978 ± 0.0010**	**0.9985 ± 0.0005**

Another important metric for evaluating the regression model is the percentage of samples whose estimation error falls within a specific range. The percentage of samples per error range for X, Y, and Z is shown in Table 6.4.

Table 6.4: Percentage of samples with estimation errors within specific ranges for X, Y, and Z Coordinates for the CNN Regression Model

Axis	**Error Range (cm)**	**Fold1(%)**	**Fold2(%)**	**Fold3(%)**	**Fold4(%)**	**Fold5(%)**	**Mean(%)**
X	$(-1, 1)$	84.95	88.91	88.87	85.22	90.51	87.69
	$(-2, 2)$	96.77	97.45	97.45	98.73	98.79	97.83
	$(-3, 3)$	98.66	99.05	98.58	99.79	99.14	99.05
	$(-4, 4)$	99.02	99.34	99.36	100.00	99.36	99.42
Y	$(-1, 1)$	90.58	90.88	92.84	93.56	96.65	92.90
	$(-2, 2)$	97.47	98.03	97.94	98.44	99.71	98.32
	$(-3, 3)$	98.66	99.93	99.43	99.65	99.93	99.52
	$(-4, 4)$	98.66	100.00	99.86	100.00	99.93	99.69
Z	$(-1, 1)$	42.69	85.56	69.43	51.41	73.11	64.44
	$(-2, 2)$	81.58	98.98	91.35	91.09	96.93	91.98
	$(-3, 3)$	94.59	99.93	96.52	99.29	99.50	97.97
	$(-4, 4)$	98.95	99.93	98.58	100.00	99.79	99.45

Figure 6.9 shows the estimation error for each data point across the five cross-

validation folds, with each fold serving once as the test set. For each coordinate, the error is the absolute difference between the predicted and true values, expressed in centimeters. Figure 6.10 shows the error for each coordinate as a function of its corresponding axis value: X error versus X, Y error versus Y, and Z error versus Z.

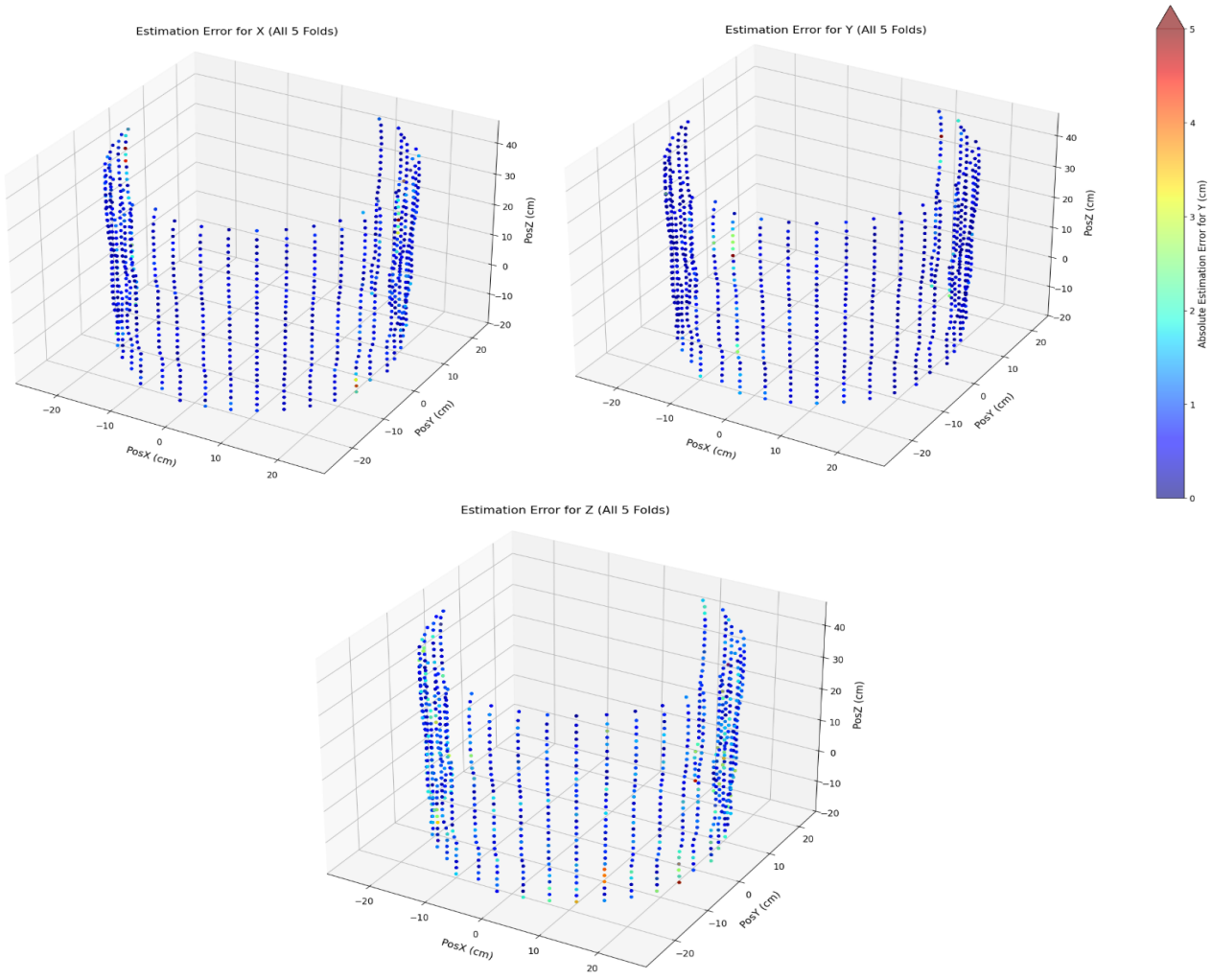


Figure 6.9: 3D scatered plot showing the absolute error value for all the points estimated in the five folds as test set for the CNN Regression Model. The results are shown for the estimation error of each coordinate in centimeters.

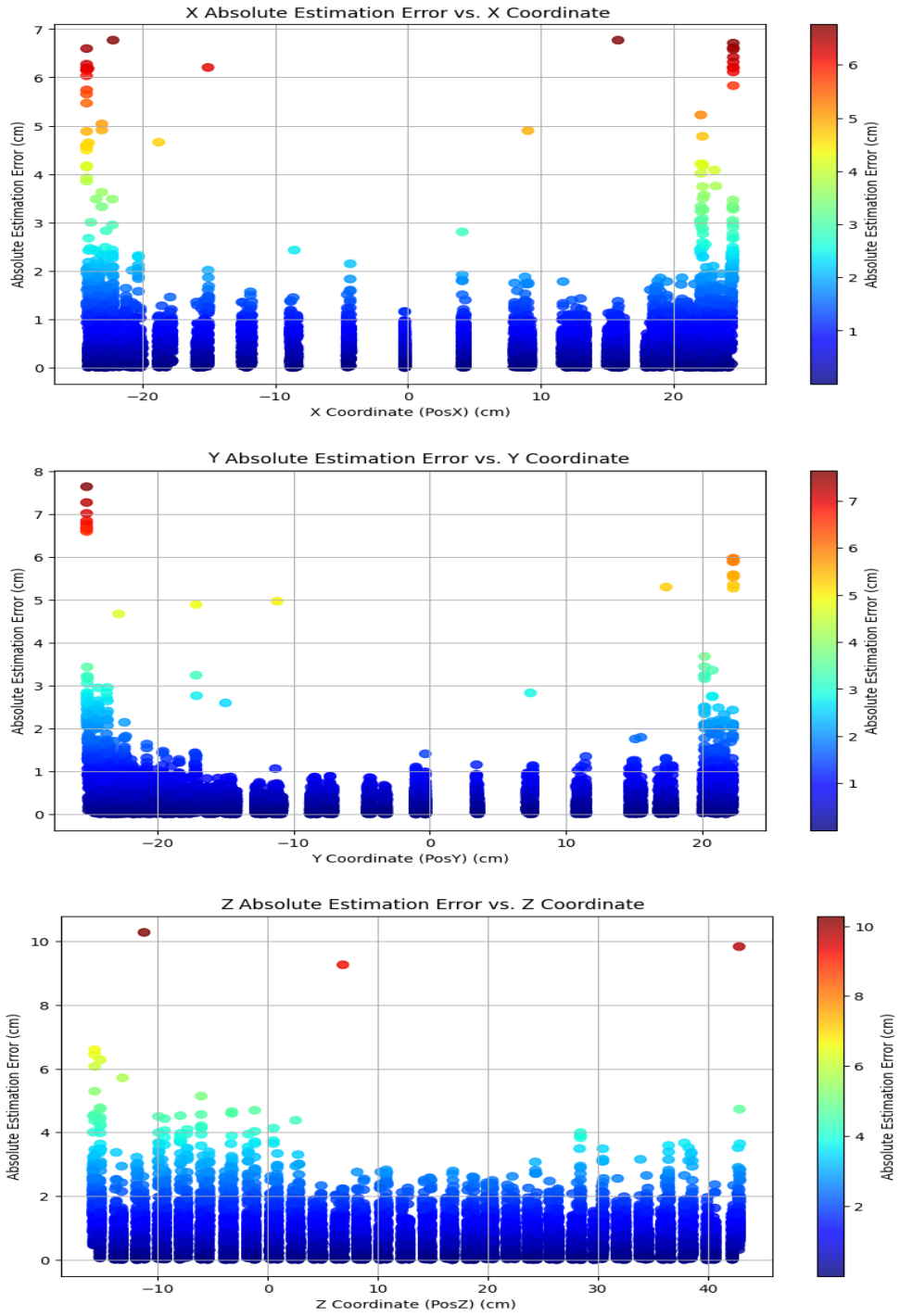


Figure 6.10: Scatered plot showing the absolute estimation error vs it respective coordinate axis for the CNN Regression Model. The results are shown for the estimation error of each coordinate in centimeters.

6.3. ESN for Acoustic Classification

The second architecture evaluated in this thesis was the multi-head Echo State Network (ESN), described in Section 5.2.2. The ESN was first evaluated as a classifier during architecture selection and hyperparameter tuning. Using the values listed in Table 5.3, the ESN was trained for 150 epochs and achieved an accuracy of 98.42%. Figure 6.11 shows the resulting confusion matrix.

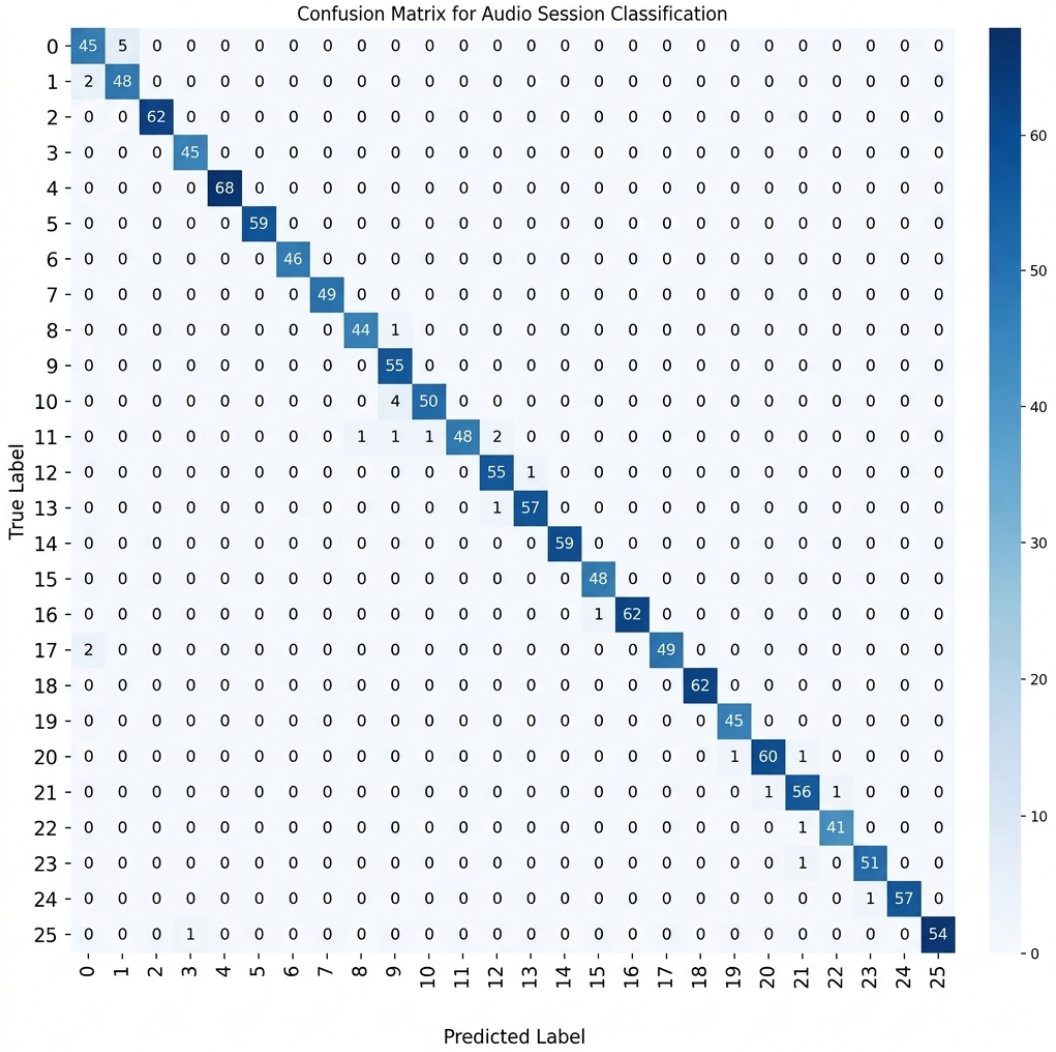


Figure 6.11: Confusion matrix for the ESN as result of the audio session classification.

6.4. ESN for Acoustic to Spatial Location Regression

As with the CNN, key performance metrics for the ESN are presented in the following section. The error-distribution histograms for the ESN regression results are shown in Figures 6.12, 6.13, and 6.14 for the X, Y, and Z coordinates, respectively.

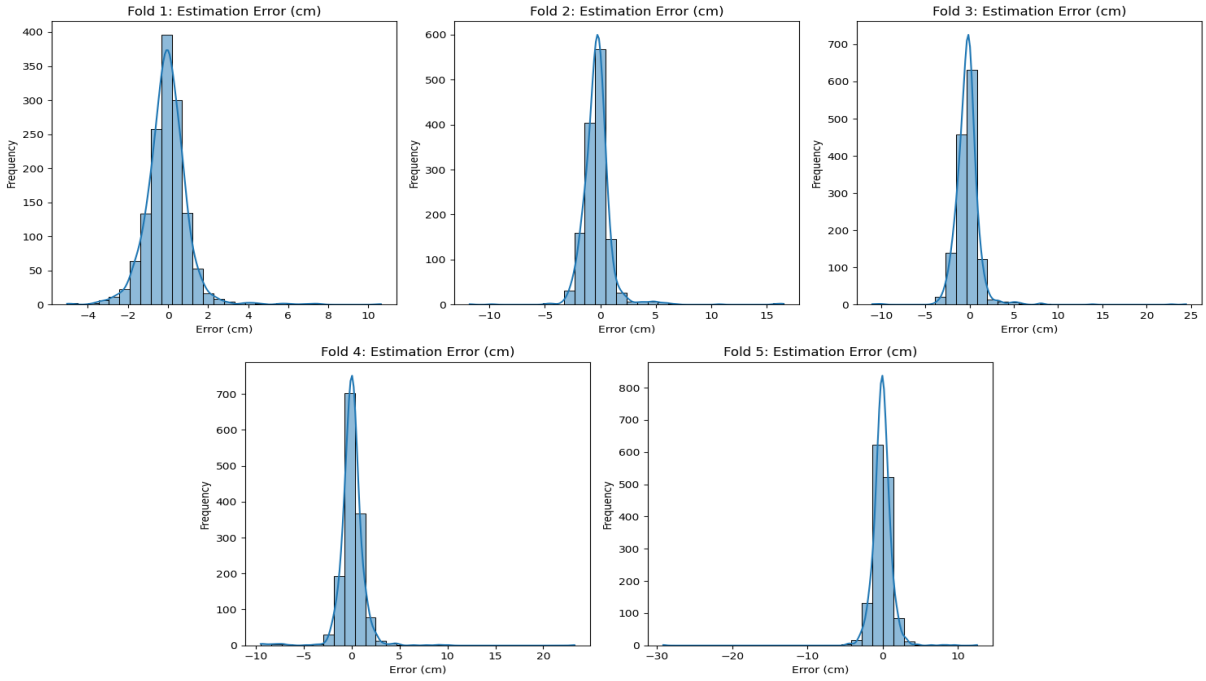


Figure 6.12: Histograms of estimation errors with ESN for X values across five folds.

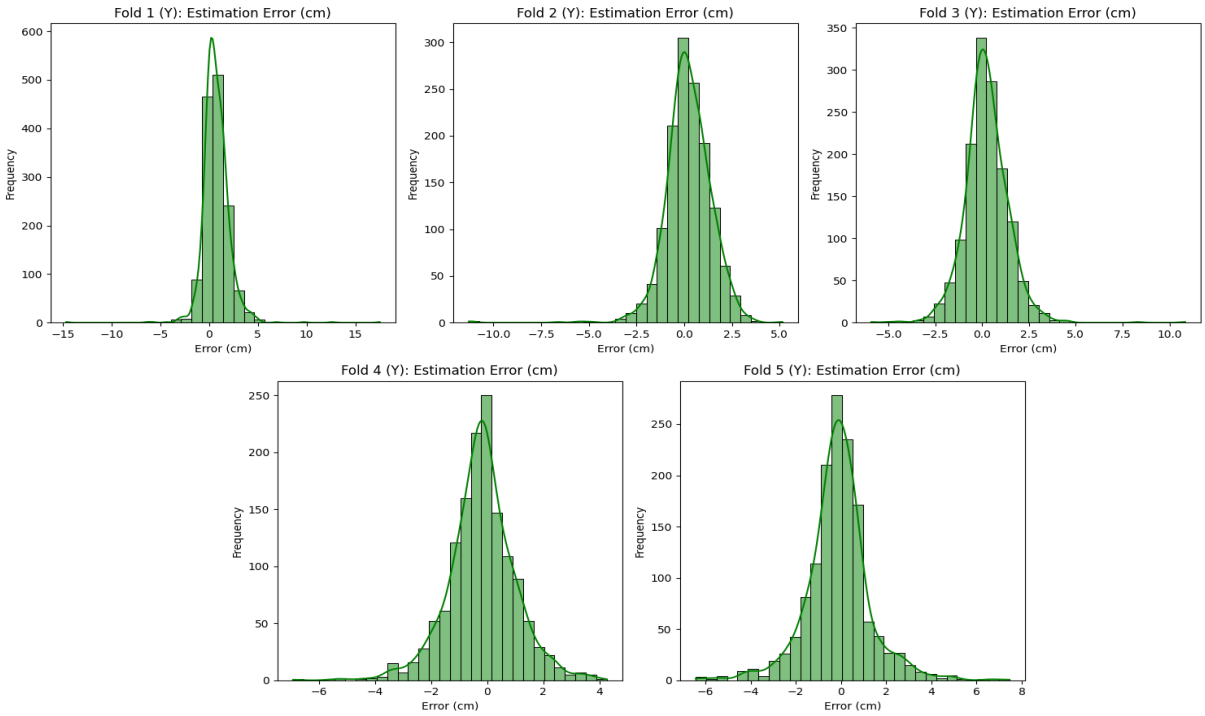


Figure 6.13: Histograms of estimation errors with ESN for Y values across five folds.

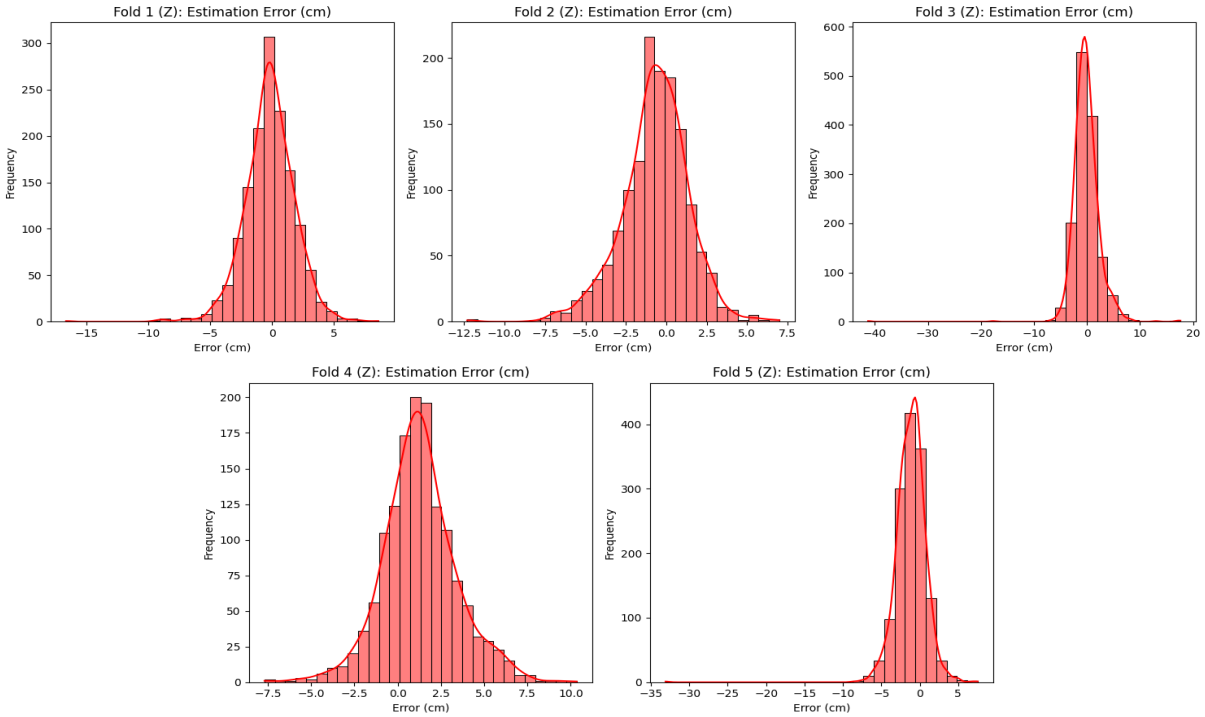


Figure 6.14: Histograms of estimation errors with ESN for Z values across five folds.

The MAE for each network is presented in table 6.5 and the Pearson correlation coefficients are presented in table 6.7.

Table 6.5: ESN Mean Absolute Error (MAE) Across Folds for X, Y, and Z Coordinates

Folds	**MAE X (cm)**	**MAE Y (cm)**	**MAE Z (cm)**
Fold 1	0.7231	1.0643	1.5111
Fold 2	0.8729	0.8977	1.6283
Fold 3	0.8955	0.8706	1.6953
Fold 4	0.7974	0.8918	1.9315
Fold 5	0.8521	0.9938	1.6847
Mean ± SD	**0.8282 ± 0.0691**	**0.9436 ± 0.0825**	**1.6902 ± 0.1535**

With respet to the distribution of errors per range, the table 6.6 show the results for the ESN evaluated in the same five fold condition as the CNN.

Table 6.6: Percentage of samples with estimation errors within specific ranges for X, Y, and Z Coordinates for the ESN across five folds

Axis	**Error Range (cm)**	**Fold1(%)**	**Fold2(%)**	**Fold3(%)**	**Fold4(%)**	**Fold5(%)**	**Mean(%)**
X	$(-1, 1)$	76.65	70.97	68.60	76.38	71.18	72.76
	$(-2, 2)$	95.01	92.20	91.85	94.13	92.58	93.15
	$(-3, 3)$	98.38	97.37	97.38	97.67	97.86	97.73
	$(-4, 4)$	99.09	98.32	98.58	98.16	99.00	98.63
Y	$(-1, 1)$	57.81	65.57	66.76	65.77	66.33	64.45
	$(-2, 2)$	87.13	91.17	91.85	90.24	86.52	89.38
	$(-3, 3)$	95.50	98.32	98.09	97.10	94.65	96.73
	$(-4, 4)$	98.10	99.49	99.36	99.43	97.50	98.78
Z	$(-1, 1)$	44.02	42.52	39.76	33.88	38.52	39.74
	$(-2, 2)$	71.87	70.17	69.17	62.52	65.98	67.94
	$(-3, 3)$	88.05	85.70	85.83	79.21	85.73	84.90
	$(-4, 4)$	95.01	92.34	92.63	89.25	93.51	92.55

The figure 6.15 shows the 3D scatter plot of the absolute errors for the ESN across the five folds and the figure 6.16 shows the 2D scatter plots of the absolute errors versus their respective coordinate axes.

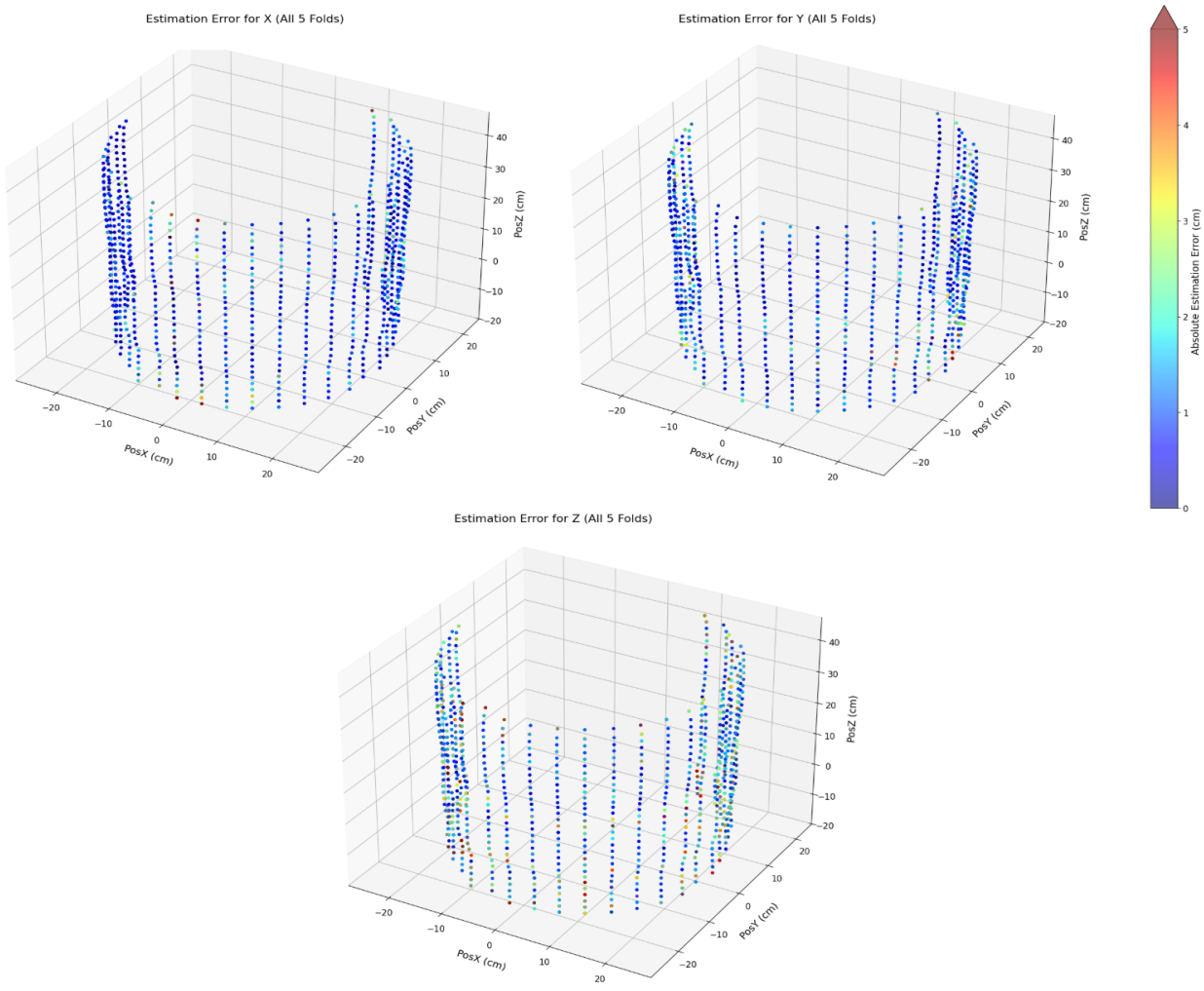


Figure 6.15: 3D scatered plot showing the absolute error value for all the points estimated with the ESN in the five folds as test set.

Table 6.7: Pearson Correlation Coefficients between Ground Truths and Predictions for the ESN across five folds

Folds	X	Y	Z
Fold 1	0.9980	0.9954	0.9929
Fold 2	0.9963	0.9966	0.9929
Fold 3	0.9958	0.9968	0.9898
Fold 4	0.9962	0.9967	0.9921
Fold 5	0.9965	0.9953	0.9936
Mean ± SD	**0.9966 ± 0.0008**	**0.9962 ± 0.0007**	**0.9923 ± 0.0015**

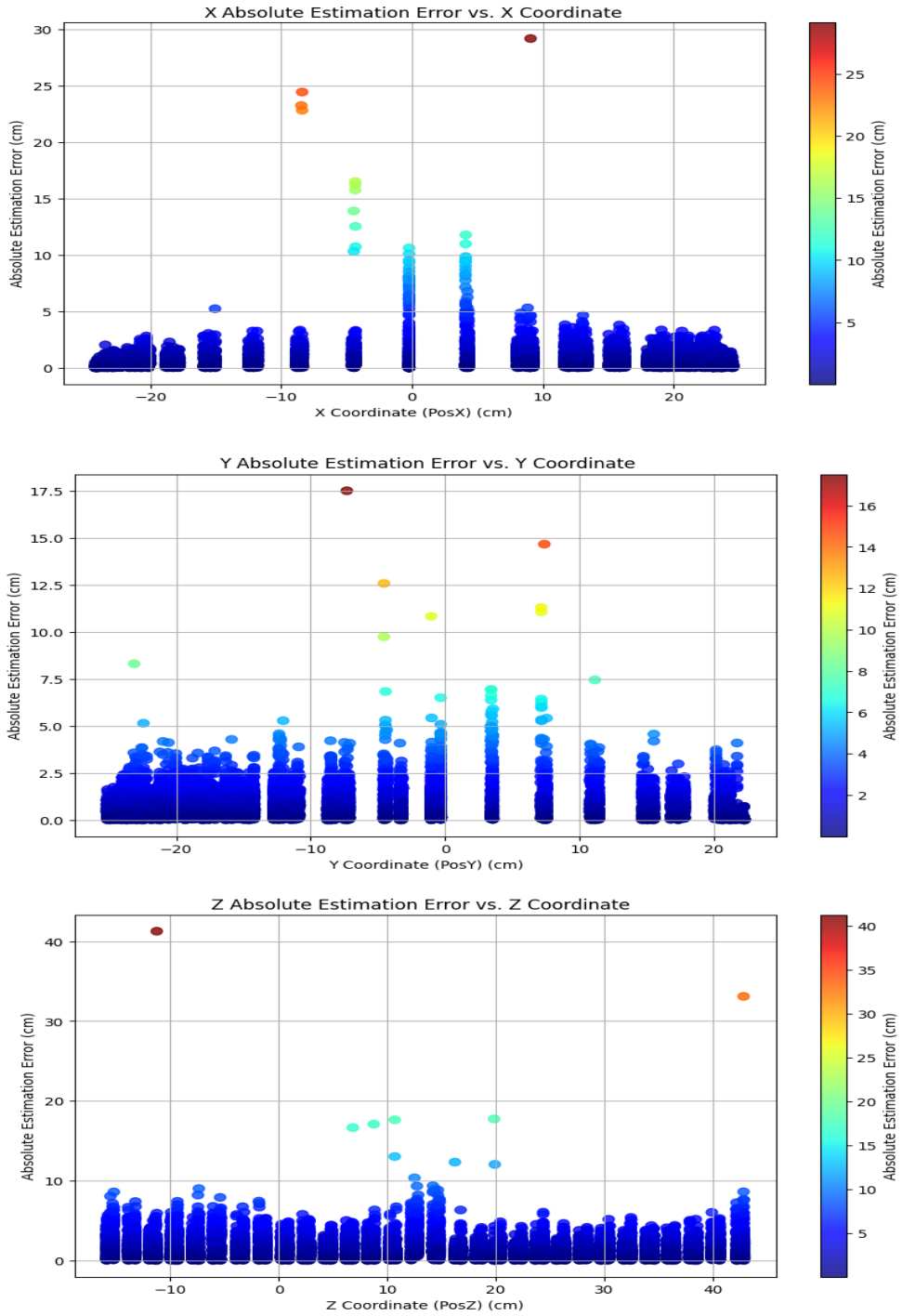


Figure 6.16: Scatered plot showing the absolute estimation error vs its respective coordinate axis for the ESN across five folds.

6.5. Beamforming

This section presents the results for the CNN and ESN models when the input signals are preprocessed with beamforming. The network architectures remain the same as previously described; the only difference is that beamforming is applied to the input signals to emphasize specific reflections according to their directions of arrival. For both models using beamforming, the Mean Absolute Error (MAE), the Pearson correlation coefficient, and absolute error distribution grahics are reported in this section. The detailed fold-wise histograms and the percentages of samples within the estimation error ranges are provided in the appendix.

6.5.1. CNN and Beamforming for Acoustic to Spatial Location Regression

For the CNN model, the input signals are preprocessed with beamforming before being converted into Log-Mel images, and the network is trained to estimate the acoustic source's 3D coordinates. The MAE for each coordinate axis across the five folds is presented in Table 6.8. The Pearson correlation coefficients for each coordinate axis are shown in Table 6.9.

Table 6.8: Mean Absolute Error (MAE) Across Folds for X, Y, and Z Coordinates for the CNN Model with Beamforming

Folds	**MAE X (cm)**	**MAE Y (cm)**	**MAE Z (cm)**
Fold 1	0.4941	0.4903	0.8064
Fold 2	0.5467	0.4418	0.7854
Fold 3	0.4246	0.4064	0.8489
Fold 4	0.4426	0.4684	0.8927
Fold 5	0.4865	0.5012	0.9552
Mean ± SD	**0.4789 ± 0.0478**	**0.4616 ± 0.0383**	**0.8577 ± 0.0683**

Table 6.9: Pearson Correlation Coefficients between Ground Truths and Predictions for the CNN Model with Beamforming

Folds	X	Y	Z
Fold 1	0.9966	0.9940	0.9980
Fold 2	0.9952	0.9952	0.9980
Fold 3	0.9966	0.9949	0.9979
Fold 4	0.9967	0.9953	0.9973
Fold 5	0.9967	0.9956	0.9962
Mean ± SD	**0.9964 ± 0.0007**	**0.9950 ± 0.0006**	**0.9975 ± 0.0008**

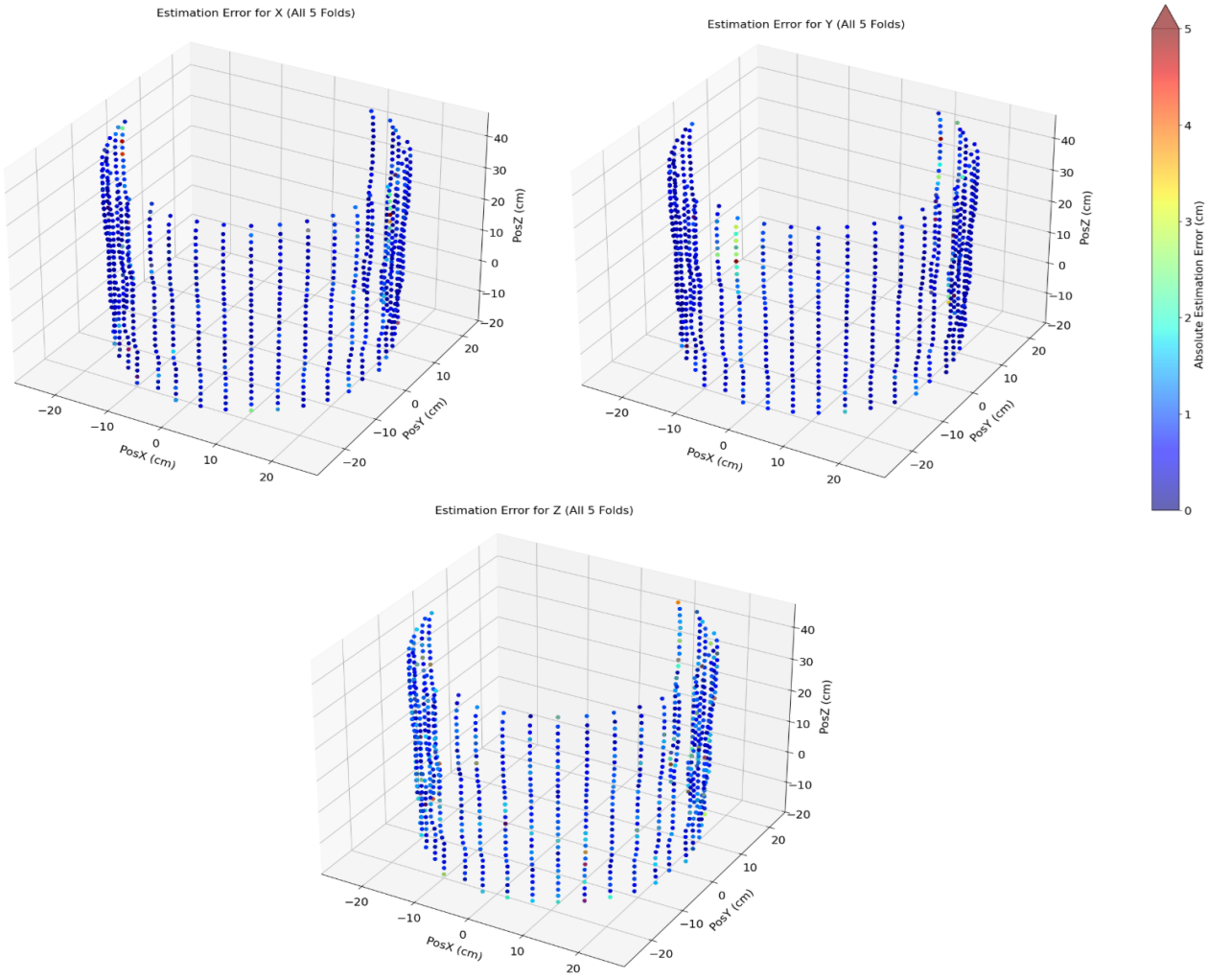


Figure 6.17: 3D scattered plot showing the absolute error value for all the points estimated in the five folds as test set for the CNN Model with Beamforming.

The figure 6.17 shows the error distribution in a 3D scattered plot of the absolute error values for all the points estimated in the five folds as the test set for the CNN Model with Beamforming. The figure 6.18 presents the absolute estimation error against the respective coordinate axis for the same model.

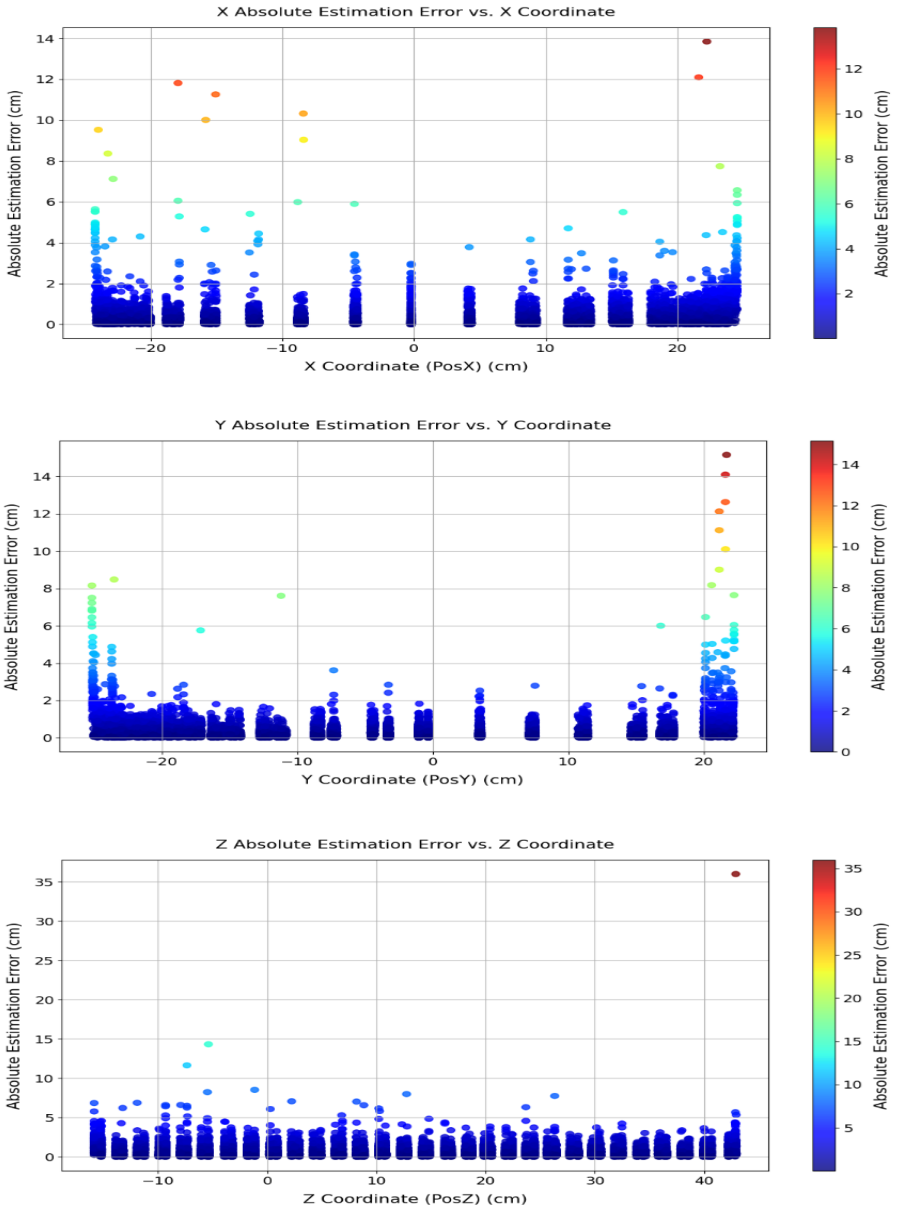


Figure 6.18: Scattered plot showing the absolute estimation error vs the respective coordinate axis for the CNN Model with Beamforming.

6.5.2. ESN and Beamforming for Acoustic to Spatial Location Regression

The last model evaluated is the Multihead Echo State Network with Beamforming. The same metrics considered in previous sections for the CNN with Beamforming are used here to evaluate this model. The main metrics include the mean absolute error in table 6.10 and the Pearson correlation coefficient for the X, Y, and Z coordinates in table 6.11.

Table 6.10: Mean Absolute Error (MAE) Across Folds for X, Y, and Z Coordinates for the ESN with Beamforming

Folds	**MAE X (cm)**	**MAE Y (cm)**	**MAE Z (cm)**
Fold 1	1.0824	1.3697	1.7033
Fold 2	1.0560	1.0145	2.3761
Fold 3	1.0589	1.1298	1.7263
Fold 4	1.2223	1.1506	1.7914
Fold 5	0.9797	1.5625	1.5734
Mean ± SD	**1.0799 ± 0.0886**	**1.2454 ± 0.2177**	**1.8341 ± 0.3149**

Table 6.11: Pearson Correlation Coefficients between Ground Truths and Predictions for the ESN with Beamforming

Folds	X	Y	Z
Fold 1	0.9872	0.9918	0.9904
Fold 2	0.9957	0.9931	0.9878
Fold 3	0.9917	0.9936	0.9894
Fold 4	0.9841	0.9954	0.9894
Fold 5	0.9935	0.9916	0.9922
Mean ± SD	**0.9904 ± 0.0047**	**0.9931 ± 0.0015**	**0.9898 ± 0.0016**

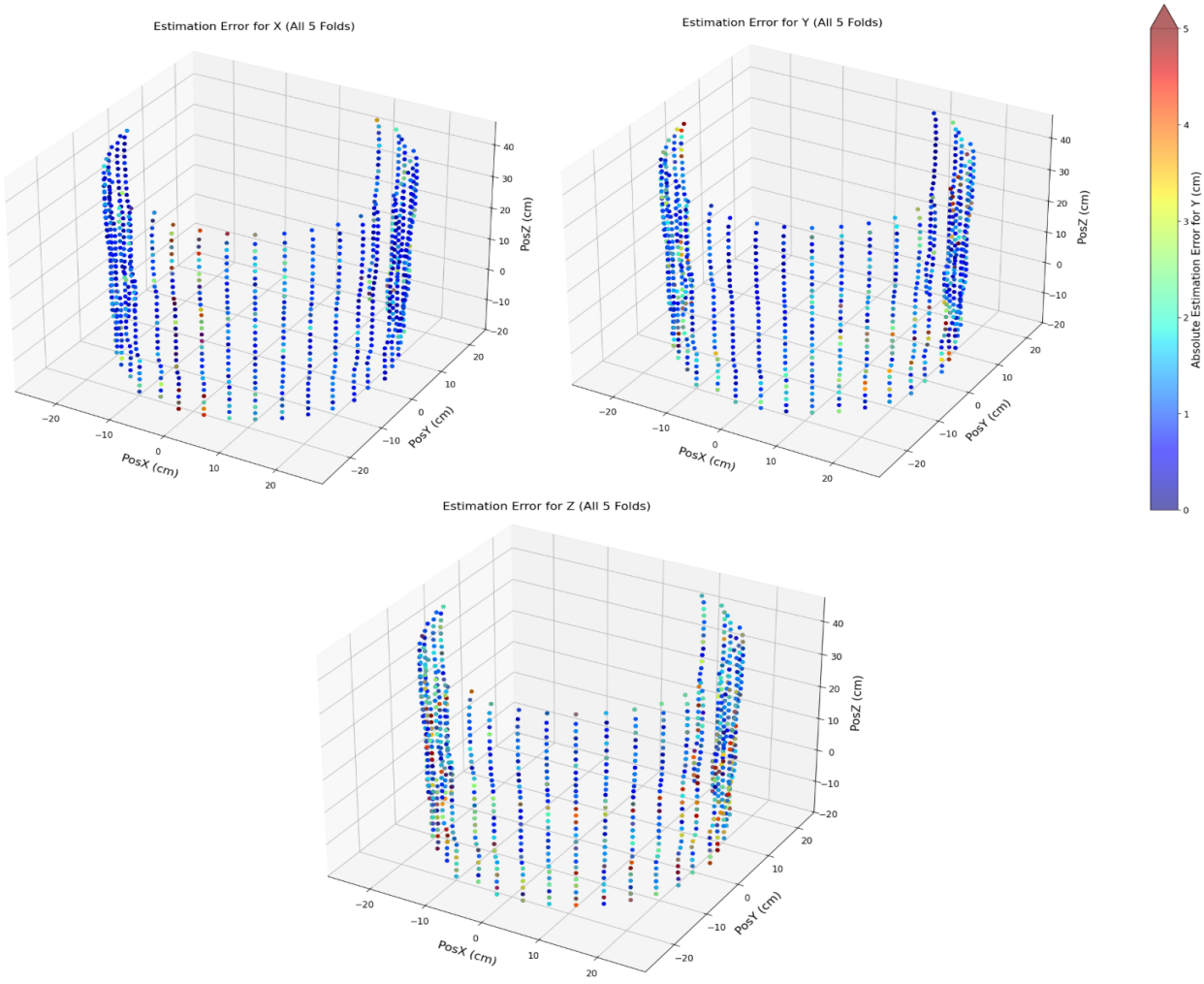


Figure 6.19: 3D scattered plot showing the absolute error value for all the points estimated in the five folds as test set for the ESN with Beamforming.

In order to visualize the distribution and the magnitude of the estimation errors under this configuration model-preprocesing setup, the scattered plots in figures 6.19 and 6.20 are presented. These plots allow visual inspection of how the errors are distributed across the spatial coordinates and help identify any patterns or biases in the model's predictions.

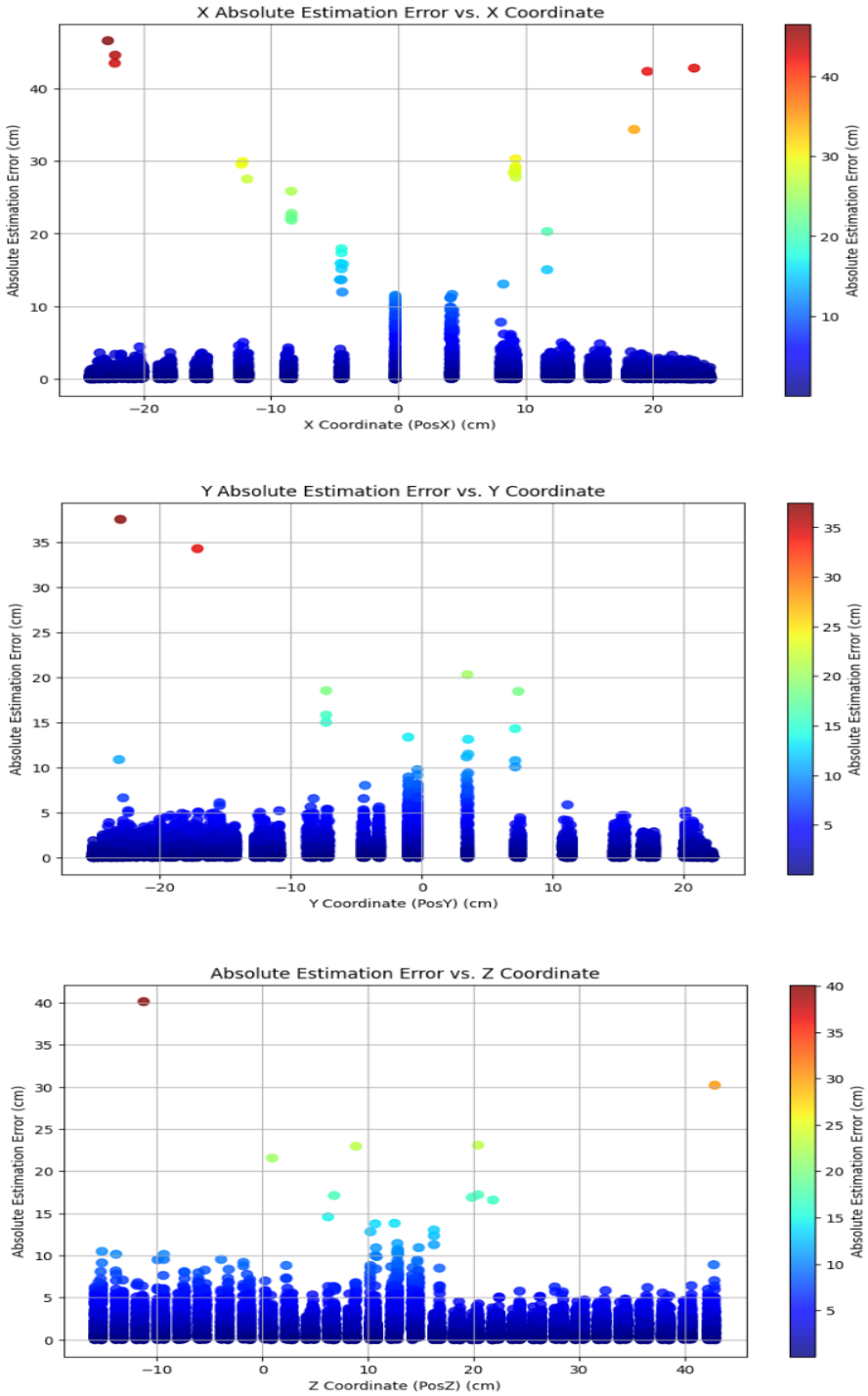


Figure 6.20: Scattered plot showing the absolute estimation error vs the respective coordinate axis for the ESN with Beamforming.

6.6. Model Comparison

To compare the models' performance in estimating the spatial coordinates, the MAE values for each fold, along with their means and standard deviations, are presented for the X, Y, and Z coordinates in Tables 6.12, 6.14, and 6.16, respectively. The fold-wise MAE values are used to rank the models for each coordinate. These rankings are then used as input to the Bayesian Plackett Luce model, whose posterior distribution is sampled using Markov chain Monte Carlo (MCMC).

The Plackett–Luce likelihood was discounted by a factor α of $(1 + K\rho)^{-1}$, with $\rho = t_{\text{test}}/t_{\text{train}} = 0.25$ and $K = 5$, resulting in a value of 0.444. This correlation penalty effectively scales the posterior probability distribution, providing a more conservative estimate of the probability that a given architecture is the best model for estimating each coordinate (X, Y, and Z).

For each coordinate, the respective MAE values and model comparison results are presented as follows:
The estimation in the X coordinate, the MAE values across the five folds for each model are presented in Table 6.12. The results from the probabilistic model comparison is presented in table 6.13. The Beam prefixed indicate the respective models incorporate beamforming into the input data processing pipeline.

Table 6.12: Mean Absolute Error Across Folds for X Coordinate Estimation for each Model

Folds	**CNN (cm)**	**ESN (cm)**	**Beam CNN (cm)**	**Beam ESN (cm)**
Fold 1	0.6019	0.7231	0.4941	1.0824
Fold 2	0.5231	0.8729	0.5467	1.0560
Fold 3	0.5569	0.8955	0.4246	1.0589
Fold 4	0.5764	0.7974	0.4426	1.2223
Fold 5	0.5062	0.8521	0.4865	0.9797
Mean ± SD	**0.5529 ± 0.0384**	**0.8282 ± 0.0687**	**0.4789 ± 0.0478**	**1.0799 ± 0.0886**

Table 6.13: Probability of Being the True Best Model (X Estimator)

Model	Probability (%)
Beam CNN	63.4
CNN	32.3
ESN	4.1
Beam ESN	0.2

For estimation in the Y coordinate, the MAE values across the five folds for each model are presented in Table 6.14. The results from the probabilistic model comparison is presented in table 6.15.

Table 6.14: Mean Absolute Error Across Folds for Y Coordinate Estimation for each Model

Folds	CNN (cm)	ESN (cm)	Beam CNN (cm)	Beam ESN (cm)
Fold 1	0.5029	1.0643	0.4903	1.3697
Fold 2	0.4467	0.8977	0.4418	1.0145
Fold 3	0.4069	0.8706	0.4064	1.1298
Fold 4	0.3836	0.8918	0.4684	1.1506
Fold 5	0.3144	0.9938	0.5012	1.5625
Mean ± SD	**0.4109 ± 0.0700**	**0.9436 ± 0.0814**	**0.4616 ± 0.0383**	**1.2454 ± 0.2177**

Table 6.15: Probability of Being the True Best Model (Y Estimator)

Model	Probability (%)
Beam CNN	52.1
CNN	42.7
ESN	5.0
Beam ESN	0.2

For estimation in the Z coordinate, the MAE values across the five folds for each model are presented in Table 6.16. The results from the probabilistic model comparison is presented in table 6.17.

Table 6.16: Mean Absolute Error Across Folds for Z Coordinate Estimation for each Model

Folds	**CNN (cm)**	**ESN (cm)**	**Beam CNN (cm)**	**Beam ESN (cm)**
Fold 1	1.3013	1.5111	0.8064	1.7033
Fold 2	0.5549	1.6283	0.7854	2.3761
Fold 3	0.8829	1.6953	0.8489	1.7263
Fold 4	1.0473	1.9315	0.8927	1.7914
Fold 5	0.7316	1.6847	0.9552	1.5734
Mean ± SD	**0.9036 ± 0.2862**	**1.6902 ± 0.1534**	**0.8577 ± 0.0683**	**1.8341 ± 0.3149**

Table 6.17: Probability of Being the True Best Model (Z Estimator)

Model	**Probability (%)**
Beam CNN	52.6
CNN	43.6
ESN	2.3
Beam ESN	1.4

Chapter 7

Discussion

7.1. CNN model and the regression problem

7.1.1. CNN model to identify cube regions and orientations

To explore the task of locating the cube inside the measurement box, we followed the problem-complexity tiers described by Vapnik in [45]. The model was first evaluated as a classifier to identify the cube's region and orientation. For the location-classification task across the acoustic sessions, the CNN achieved 98.59% accuracy on the test set. This result indicates that the audio signals, once transformed into spectrograms, contain sufficient information for spatial localization and that the network can extract the features needed to identify the cube's region within the measurement box. The confusion matrix in Figure 6.1 shows a clear diagonal pattern, indicating strong classification performance. Twenty of the twenty-six classes achieved 100% recall. Most errors occurred between neighboring classes. Class 1 had the most misclassifications, with 13 samples classified as class 0, resulting in 75% accuracy for class 1. This result may be explained by the small distance between the points corresponding to these two sessions, as shown in Figure 6.4 where S1 corresponds to class 0, and S2 corresponds to class 1.

In the orientation-classification task, the network was required to assign the cube's orientation to one of nine classes. The CNN achieved 29.06% accuracy on the test set. Unlike the location-classification results, the confusion matrix in Figure 6.2 does not show a clear diagonal pattern. Instead, it reveals three clusters of misclassifications near the diagonal. In the first cluster, samples from classes 0, 1, and 2 are often classified as class 1. In the middle cluster, samples from classes 2, 3, 4, and 5 are often classified as class 4. In the rightmost cluster, samples from classes 5, 6, 7, and 8 are often classified as class 6. Overall, predictions are concentrated in classes 1 (10 degrees), 4 (40 degrees), and 6 (60 degrees). This pattern suggests that the network tends to favor a few broad orientation categories spaced approximately 20 to 30 degrees apart rather than reliably

distinguishing all nine classes, which are spaced at 10-degree increments.

To further investigate whether the cube's proximity to the microphones and the orientation of its front face relative to the sides of the measurement box affected classification performance, the data were divided into three regions. The regional accuracy results in Table 6.1 show no improvement when the analysis is restricted to these regions. The even lower accuracy may be explained by the reduced number of training samples in each region. Although accuracy remained low in this regional classification task, the confusion matrices in Figure 6.3 show a more pronounced concentration of samples along the diagonal.

The cross-correlation plots in Figure 6.5 show that as the Euclidean distance between two locations increases, the normalized cross-correlation decreases, indicating that the acoustic signals become less similarity as the cube moves farther from the reference point. By contrast, changes in orientation do not produce a consistent decrease in cross-correlation; in some cases, it increases, indicating that the signals at certain orientations remain similar to the reference signal. This similarity may partly explain the CNN's poor performance on the orientation-classification task.

The two tasks show a pronounced difference in performance: spatial-location classification achieved over 98% accuracy, whereas orientation classification achieved only 29.06%. This difference may arise because rotating the cube changes the orientation of its surfaces without substantially changing their positions relative to the microphones and the wall in the box. Consequently, orientation may be encoded in subtle inter-channel phase shifts or in the attenuation of certain echoes, rather than in pronounced Time-Difference-Of-Arrival (TDOA) cues. By contrast, translating the cube changes the source–microphone distances and may produce more distinct TDOA cues and changes in the reverberation pattern. The CNN may therefore have difficulty learning orientation-dependent features in the spectrograms, which could explain why its predictions are concentrated in a few orientation classes.

7.1.2. CNN model to estimate the cube location

For regression, the CNN was used to estimate the cube's X, Y, and Z coordinates. Figures 6.6, 6.7, and 6.8 show the prediction-error histograms for the three coordinates across the five folds. The histograms indicate that the model generally localizes the cube with reasonable accuracy: most errors fall within approximately ±2 cm, a relatively narrow range compared with the dimensions of the measurement box and the cube. Most errors are concentrated around zero, suggesting that the network's predictions are close to the true positions. However, some larger errors occur, possibly because of challenging acoustic conditions, such as recording noise. The X and Y histograms show similar results. For Z, the distributions for folds 1 and 4 are broader, with more errors exceeding ±2 cm, suggesting that the network has greater difficulty estimating this coordinate. Overall, the histograms qualitatively demonstrate the CNN's strong spatial-localization capability, consistent with the high classification accuracy reported in the previous section. The error distributions are approximately bell-shaped and centered around zero, indicating good estimation performance across all three dimensions.

In this work, the MAE is the primary figure of merit used to evaluate a regression model's estimation performance [34]. Table 6.2 summarizes the mean and standard deviation of the MAE for the X, Y, and Z coordinates across the five folds. The CNN achieves the lowest MAE for the Y coordinate, followed by the X coordinate. Although the X-coordinate MAE is slightly higher than the Y-coordinate MAE, the values are similar, at approximately 0.5 cm. For the Z coordinate, the mean MAE is nearly twice as high, at 0.9 cm, and the standard deviation is larger. These results are consistent with the histograms and further support the conclusion that the network estimates the cube's horizontal position more accurately than its vertical position.

Another metric used to evaluate regression performance is the Pearson correlation coefficient [34]. Table 6.3 reports this coefficient for all the folds and coordinates. The coefficients are consistently very close to 1, indicating a strong linear relationship

between the predicted and true coordinates. These results further support the network's ability to estimate the cube's position in the measurement box.

To analyze the error distribution across the specified ranges, Table 6.4 presents the percentage of samples whose estimation errors fall within each range. The results are consistent with the previous paragraph: Y is the most accurately estimated coordinate, with 92.90% of samples falling within ±1 cm and 98.32% within ±2 cm, averaged across the five folds. X ranks second, with 87.69% of samples falling within ±1 cm and 97.83% within ±2 cm, comparable to Y. Z has the largest errors and the greatest variation across folds: only 64.44% of samples fall within ±1 cm. The proportion within ±3 cm rises to 98.32%, meaning that 8.92% of Z-coordinate errors exceed ±2 cm. For all three coordinates, fewer than 0.6% of samples have errors exceeding ±4 cm; equivalently, the proportions with errors within ±4 cm are 99.42% for X, 99.69% for Y, and 99.45% for Z.

The error distributions across the three dimensions can be observed in Figure 6.9, which shows the X, Y, and Z errors throughout the measurement box. The visualization indicates that errors are distributed relatively uniformly across the measurement space. Most errors are close to zero, while larger errors occur in a few regions; however, no clear clustering pattern is apparent. Figure 6.10 shows the error for each coordinate as a function of its corresponding coordinate value. For X, the absolute estimation error remains low near the center and increases toward the edges, where most outliers occur. A similar pattern is observed for Y, although the errors are lower than those observed for X. This suggests that the network estimates the Y coordinate more accurately than the X coordinate near the center. This result is notable because the network receives binaural input for X estimation, which could allow it to use the physical separation between the two microphones to learn interaural time differences. Despite this, Y estimation performs better. One possible explanation is that the cube's position along the Y axis produces distinctive interference patterns at different distances from the

speaker and microphones. These changes in distance relative to the microphones may affect both the echo patterns and their magnitudes, particularly when the cube is near the speaker or the microphones. Such effects may help explain why Y estimation errors are lower than X estimation errors. The errors for Z are higher than those for X and Y. The Z-coordinate estimates may be affected by the uneven distribution of values around zero: the dataset contains more positive than negative values. Consequently, the network is trained on more positive samples than negative ones. This imbalance may contribute to the larger estimation errors for negative Z values. In addition, the robotic arm's base is located in a side (the base of the measurement box) perpendicular to the Z axis. Because this area is one of the two main surfaces contributing to binaural reverberation, it may complicate estimation in that part of the measurement space.

7.2. ESN model and the regression problem

Similar to the analysis for the CNN network, the ESN was first evaluated as a classifier to assess the ability to extract the acoustic features needed to locate the cube within the measurement box. The ESN achieved 98.42% accuracy. This result is 0.17% lower than the CNN's 98.59% demostrating that in clasification the ESN had a equate acuracy to the CNN. The confusion matrix in Figure 6.11 shows a strong diagonal, indicating close agreement between the true and predicted labels. Most misclassifications occur between neighboring classes. Unlike the CNN, the ESN classifies sessions 0 and 1 more accurately.

Figures 6.12, 6.13, and 6.14 show the ESN error distributions for the X, Y, and Z coordinates, respectively. As with the CNN, these distributions are approximately bell-shaped and centered around zero. However, the ESN distributions are wider, indicating greater variability than the CNN distributions.

Overall, the ESN's MAE values in Table 6.5 are higher than the CNN's values in

Table 6.2. Another difference is that the relative performance of the X and Y estimators is reversed: the ESN achieves a slightly lower MAE for X than for Y, while Z has the highest MAE of the three coordinates. The ESN hyperparameters were initially tuned using the X coordinate. Although X and Y have similar input-coordinate distributions and use the same ground-truth label transformation, applying the X-optimized hyperparameters to Y may have contributed to its higher MAE. The Z-coordinate data also have a different spatial distribution, and their ground-truth values do not require the arcsine normalization described in Equation 5.2. These differences may further affected how well the selected hyperparameters generalize across coordinates.

The error percentages across the tolerance ranges in Table 6.6 are consistent with the wider ESN error distributions shown in the histograms. A tolerance of ± 2 cm encompasses a smaller proportion of the estimates, whereas a tolerance of ± 3 cm is needed to include approximately 97% of the samples.

The Pearson correlation coefficients on table 6.7 also exceed 0.99 in most cases. The only exception is the Z estimator in fold 3 with a value of 0.98, suggesting that the regression model may have difficulty capturing the relationship between the features and target values for some samples in this fold.

Figure 6.15 shows the spatial distribution of estimation errors across all three dimensions. The scatter plot suggests that the errors are distributed relatively uniformly throughout the measurement region. However, the coordinate-wise plots in Figure 6.16 indicate that the largest errors occur in a localized central region rather than near the limits of the measurement space. This pattern may be explained by the multi-head architecture used for coordinate estimation. Two regression heads estimate the cube's position in the negative and positive coordinate ranges, respectively, while a third head selects which estimate to use. This selection may be less reliable near the center, where the two ranges meet. A fourth head identifies outliers by comparing the

estimates; however, if their discrepancy is below the specified threshold, an incorrect head selection may not be flagged. This could explain why the central region remains prone to high errors. Training both regression heads on an overlapping region around the boundary could mitigate this issue: each head would then learn from samples on both sides of the boundary, reducing errors when the selector chooses the inappropriate head.

7.3. Beamforming

This section discusses the performance of the CNN and Echo State Network (ESN) models after beamforming was applied to the input signals.

7.3.1. CNN with Beamforming

Comparing the MAE values for the CNN with beamforming in Table 6.8 with those for the CNN without beamforming in Table 6.2, we observe that beamforming yields lower MAE values for the X and Z coordinates. For the Y coordinate, the CNN without beamforming achieves a lower mean MAE. The Pearson correlation coefficients in Table 6.9 remain above 0.99 for most coordinates, indicating that the CNN with beamforming maintains a strong linear relationship between the predicted and ground-truth positions.

The population samples percentages across the tolerance ranges in Table 9.1 indicate that beamforming improves the proportion of estimates within tighter tolerances for the X and Z coordinates. For the Y coordinate, the improvement is less pronounced, consistent with the MAE results.

Figure 6.17 shows the spatial distribution of estimation errors across the three coordinates for the CNN with beamforming. As in the case without beamforming, the errors appear to be distributed relatively uniformly across the measurement region. The coordinate-wise plots in Figure 6.18 show that, for X and Z, outliers are less

frequent near the edges than in the corresponding plots without beamforming in Figure 6.10. This pattern is not observed for Y, where errors occur near the edges in both cases. This difference may help explain why beamforming reduces the MAE for X and Z but not for Y. This result is expected because the binaural input signals provide spatial cues that support beamforming for X and Z estimation, but not for Y estimation.

7.3.2. ESN with Beamforming

In the case of the ESN with beamforming, the MAE values in Table 6.10 compared with those for the ESN without beamforming in Table 6.5, are higher for all the three coordinates, indicating that beamforming negatively impacts the ESN's estimation accuracy across the board.

The population percentages across the tolerance ranges in Table 9.2 are more spread out compared to the ESN without beamforming, indicating that fewer estimates fall within tighter tolerances. This observation aligns with the higher MAE values, confirming that beamforming negatively affects the ESN's estimation accuracy.

The Pearson correlation coefficients in Table 6.11 exceed 0.99 in most cases, suggesting that the ESN with beamforming generally maintains a strong linear relationship between the predicted and ground-truth positions. However, four fold-coordinate combinations have coefficients below 0.99, compared with only one for the ESN without beamforming in Table 6.7. This indicates that beamforming may weaken the correlation for some folds and coordinates.

Figure 6.19 shows the spatial distribution of estimation errors for the ESN with beamforming. As in the previous cases, the errors are distributed relatively uniformly across the measurement region without evidence of clustering. The coordinate-wise plots in Figure 6.20 indicate that the largest errors remain concentrated in the central region of the measurement area. This suggests that beamforming does not substantially

change the spatial pattern of the ESN's errors, although more outliers appear than in the plots compared with the ESN without beamforming in Figure 6.16.

7.4. Model Comparison

The MAE values for the different models and configurations are summarized in Tables 6.12, 6.14, and 6.16. this tables are used define tthe likehood function of each model and probide the potential function to sample the distribution of the bayesian model employ to compare the performance of the models across the different coordinates.

The first comparison is for the X coordinate, and is presented in Table Table 6.13. The CNN model with bemforming had the highest probability with a value of 63.4% of being the best model according to the Bayesian Plackett-Luce analysis. The second highest probability is attributed to the CNN model without beamforming. This results is consistent with the MAE values in Table 6.12, where the Beam CNN exhibits the lowest mean error, followed by the CNN without beamforming. The ESN models, both with and without beamforming, have higher MAE values and correspondingly lower probabilities of being the best model. The Beamforming for the case of the ESN appears to negatively impact its performance, as indicated by the higher MAE and lower probability compared to the ESN without beamforming.

For the Y coordinate, the comparison is presented in Table 6.15. The Beam CNN again has the highest probability with a value of 52.1% of being the best model, followed by the CNN without beamforming. This is not consistent with the mean values presented in Table 6.14, where the CNN without beamforming exhibits the lowest mean error, followed by the Beam CNN, but the Bayesian model assigns a higher probability to the Beam CNN due the count of folds where beam CNN performs bettern than the CNN (3:2) and the magnitude of the diferences in those

folds. In the case of the ESN models, both with and without beamforming, the probabilities remain low, reflecting their higher MAE values and the negative impact of beamforming on the ESN's performance for the Y coordinate as well.

The comparison of models in Z keep the trending pattern observed for the X and Y coordinates. The Beam CNN again demonstrates superior performance with the highest probability of being the best model, followed by the CNN without beamforming. The ESN models, particularly the Beam ESN, continue to show lower probabilities. The results in probabilities correspond to the ranking implied by the values mean MAEvalues presented in Table 6.16.

Overall, the Beam CNN consistently demonstrates the highest probability of being the best model across all three coordinates, indicating its performance could be considered the top performer in spatial coordinate estimation. The CNN without beamforming generally follows as the second-best model, while the ESN models, particularly the Beam ESN, show lower probabilities. The probability values in the tables are conservative due the small diferences in MAE values among the top-performing models and the fold cross-validation penalty applied in the Bayesian Plackett-Luce analysis.

The fact that Beamforming only improves the performance of CNN model could be explained by the effect of beamforming discused in secction 6.5.1 that the frequency of big errors on the edges of the measurement area is reduced, the principle of use beamforming is to highlight the coustic echoes originated from the edges of the measurement area and as discused from the observation of figure 6.10, these regions in the edges are the ones with bigger errors, which benefits the CNN models. The ESN model, on the other hand, do not benefit as much from this effect as is show in figure 6.16 the errors in the edges are not the main problem for this model. and the overall negative impact of beamforming on the ESN could be attributed to parameters tuning where set suboptimally for the beamformed input signals.

Although beamforming improves the CNN's performance and degrades the ESN's, both models without beamforming achieve MAE values below 2 cm for all three coordinates (Tables 6.12, 6.14, and 6.16). These results indicate that both models perform reasonably well. Despite its higher MAE values, the ESN maintains competitive accuracy while using only 3% as many trainable parameters as the ResNet-18 CNN. This makes the ESN a more parameter-efficient option for regression using acoustic features, which may be advantageous when computational resources are limited or real-time processing is required. Overall, while the CNN achieves lower absolute errors, the ESN offers a compelling trade-off between accuracy and model complexity.

Chapter 8

Conclusions and Future Work

This chapter presents the final conclusions obtained through the research effort and the future work that can be derived.

8.1. Conclusions

- The CNN and ESN architectures are capable to learn from acoustic representations like Log-Mel spectograms the mapping function from audio features to the target coordinates that define the location of an cube of 20cm length inside an enclosed espace of $1m^3$ with a MAE less than 2cm as clasification and regression tasks.

- The orientation of the cube can not be accurately determined using the current acoustic representations and network architectures.

- CNN model had the biggest errors located in the edges of the measurement space for coordinates X, Y and Z compare to the ESN that had the errors with high magnitude in the center region.

- Without beamforming CNN was the best overall model according to the Bayesian comparison, while the ESN remained a competitive and more parameter-efficient alternative, using approximately 3% as many trainable parameters as ResNet-18 CNN.

- Beamforming improved CNN performance, especially for the X and Z coordinates, but did not consistently improve Y estimation and negatively affected the ESN.

- The beamforming CNN had the higest probability of being the best model according to the Bayesian comparison, followed by the non-beamforming CNN and then the ESN.

8.2. Future Work

Future work reusing the current dataset could include:

- Investigate the efect of filtering the common echoes from the acoustic representations, the echoes that are present in all measurements and do not carry information about the cube's location. This preprocessing step could speed up the training process and potentially improve the localization accuracy by reducing irrelevant information.
- For the Multi head ESN architecture, explore the process of training the negative and positive feedforward networks with overlap to reduce the errors in the transition region.
- Explore the idea to use a multihead aproach with the CNN to see if the estimation can be improve.
- Analize the audio of the big error outliers to identify if exist especific patterns in the signal that can be filter to remove any noise that affect the estination.
- Investigate the data aumentation thecniques to balance the dataset and improve the generalization of the models.

Chapter 9

Appendixes

9.1. Source Code

Pointer or description of the source code can be given in this section..

9.2. Gathered Data from the Experiments

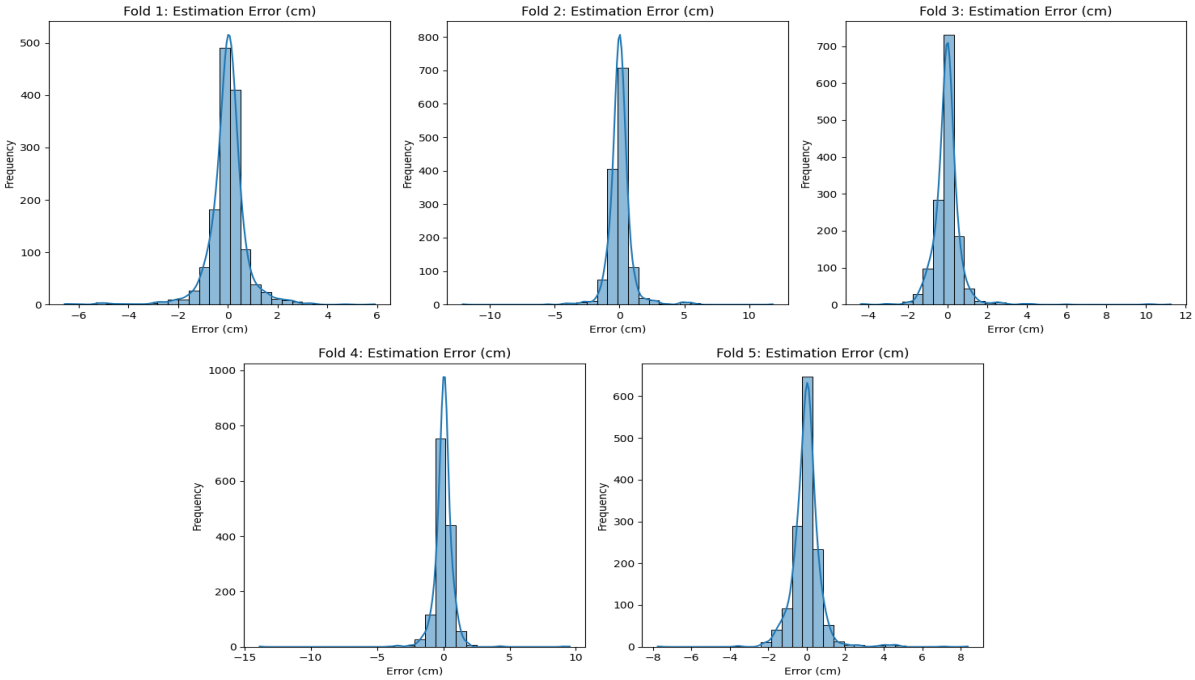


Figure 9.1: Histograms of estimation errors with Beamforming and CNN for X values across five folds.

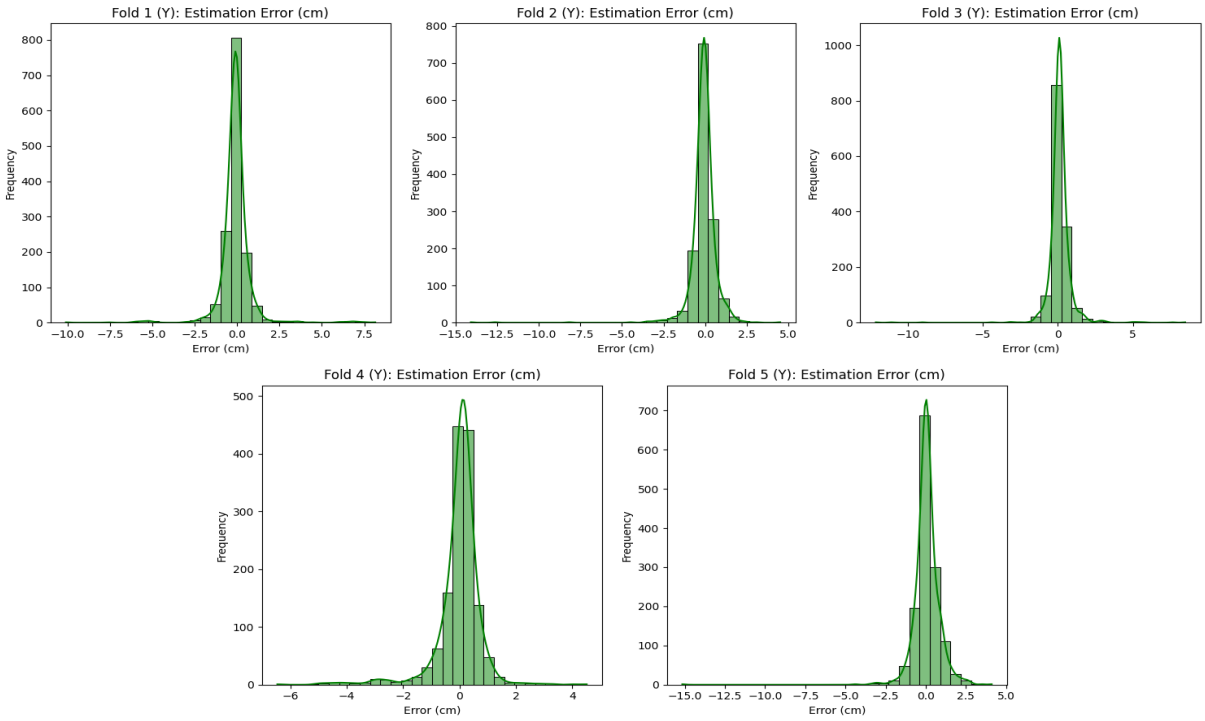


Figure 9.2: Histograms of estimation errors with Beamforming and CNN for Y values across five folds.

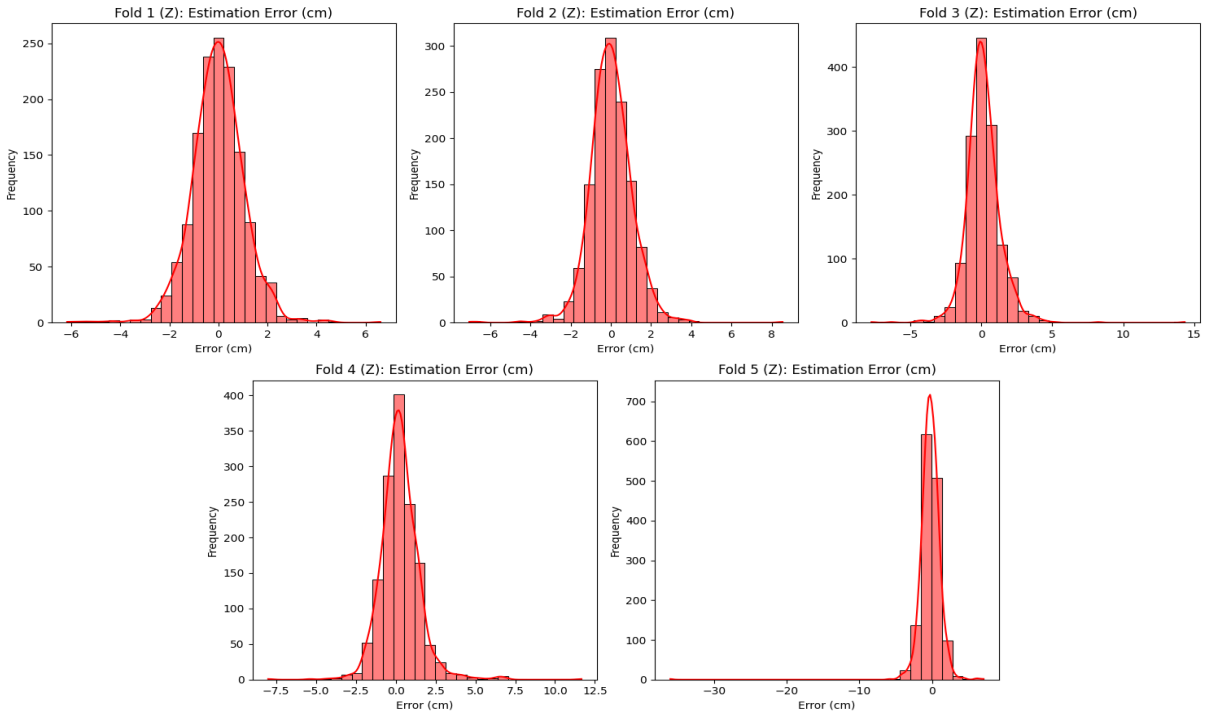


Figure 9.3: Histograms of estimation errors with Beamforming and CNN for Z values across five folds.

Table 9.1: Percentage of samples with estimation errors within specific ranges for X, Y, and Z Coordinates

Axis	Error Range (cm)	Fold1(%)	Fold2(%)	Fold3(%)	Fold4(%)	Fold5(%)	Mean(%)
X	$(-1, 1)$	87.97	87.02	91.00	90.74	88.16	88.98
	$(-2, 2)$	96.20	96.21	98.23	98.80	97.79	97.44
	$(-3, 3)$	98.66	98.25	99.08	99.29	98.86	98.83
	$(-4, 4)$	99.16	98.69	99.43	99.58	99.22	99.21
Y	$(-1, 1)$	90.08	90.66	92.77	90.17	86.95	90.13
	$(-2, 2)$	96.98	98.18	98.37	96.46	97.50	97.50
	$(-3, 3)$	98.17	99.27	98.94	98.23	99.50	98.82
	$(-4, 4)$	98.45	99.64	99.29	99.08	99.79	99.25
Z	$(-1, 1)$	70.75	72.36	70.45	66.62	63.38	68.71
	$(-2, 2)$	93.53	94.60	91.78	92.72	90.58	92.64
	$(-3, 3)$	98.59	98.47	97.52	97.31	97.43	97.86
	$(-4, 4)$	99.23	99.64	99.08	98.59	98.93	99.09

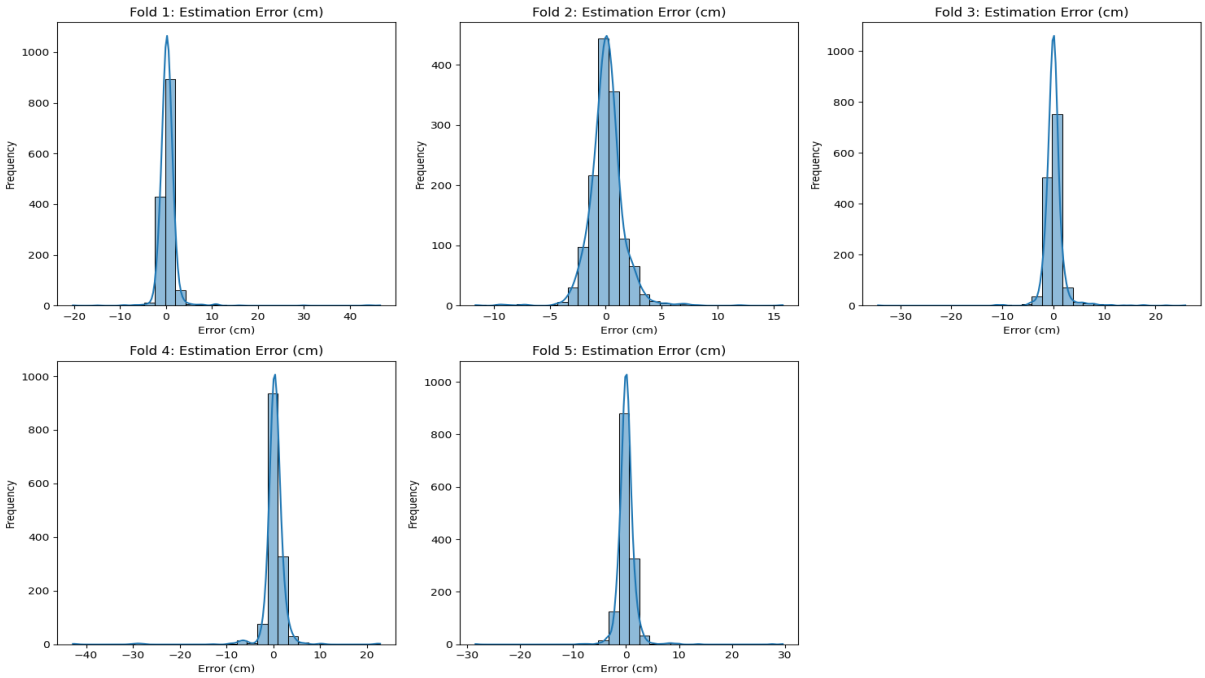


Figure 9.4: Histograms of estimation errors with Beamforming and ESN for X values across five folds.

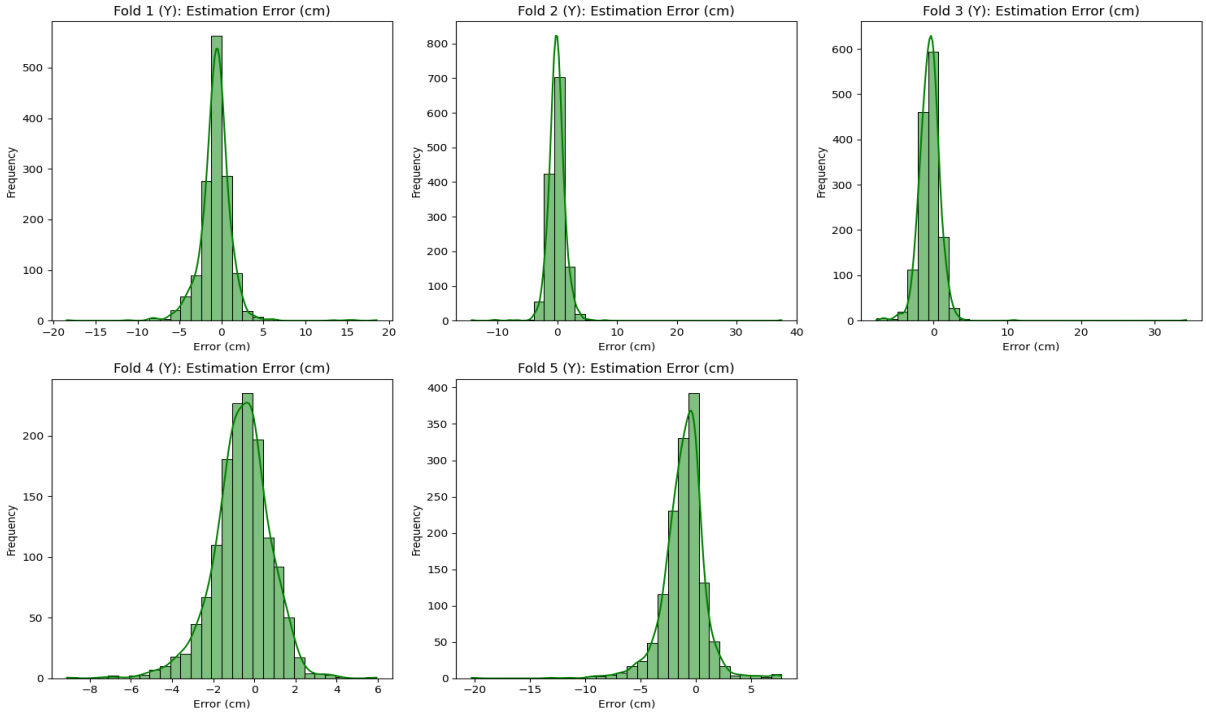


Figure 9.5: Histograms of estimation errors with Beamforming and ESN for Y values across five folds.

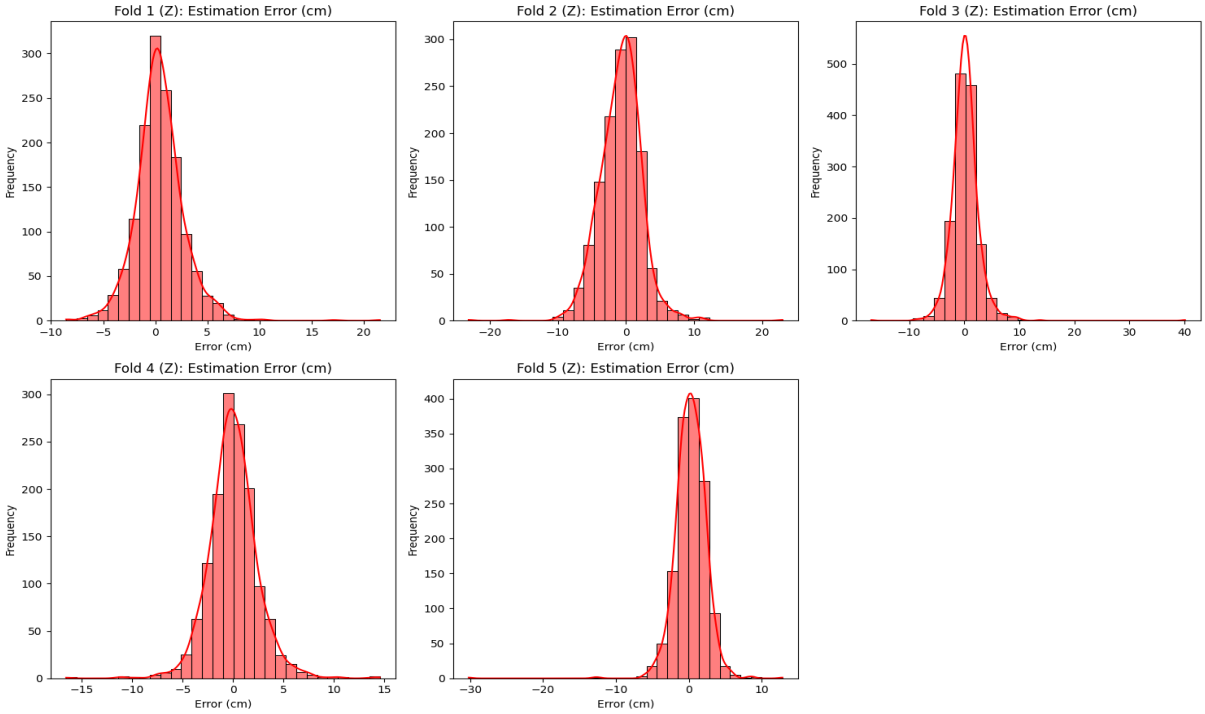


Figure 9.6: Histograms of estimation errors with Beamforming and ESN for Z values across five folds.

Table 9.2: Percentage of samples with estimation errors within specific ranges for X, Y, and Z Coordinates

Axis	Error Range (cm)	Fold1(%)	Fold2(%)	Fold3(%)	Fold4(%)	Fold5(%)	Mean(%)
X	$(-1, 1)$	65.75	63.02	69.10	65.98	68.26	66.42
	$(-2, 2)$	91.84	86.00	89.37	87.34	90.51	89.01
	$(-3, 3)$	96.55	95.55	95.25	94.20	96.15	95.54
	$(-4, 4)$	97.82	97.96	96.67	96.32	98.22	97.40
Y	$(-1, 1)$	50.98	63.31	55.42	53.61	44.79	53.62
	$(-2, 2)$	79.47	88.26	85.68	84.94	72.40	82.15
	$(-3, 3)$	90.08	96.50	96.46	94.41	87.73	93.04
	$(-4, 4)$	94.80	98.98	98.37	98.02	93.51	96.73
Z	$(-1, 1)$	41.63	29.83	41.53	39.11	40.23	38.47
	$(-2, 2)$	68.78	54.27	70.16	66.69	70.97	66.17
	$(-3, 3)$	83.54	70.97	83.91	82.39	87.95	81.75
	$(-4, 4)$	91.49	81.77	91.92	91.51	95.44	90.43

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